

# Stochastic Simulation of the UK's Disjointed International Teacher Recruitment: A Model of the Vicious Cycle

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- [Accounting conventions \(spend vs waste; year-1 horizon\)](#)

```
# -----  
# Block 1 – DfE intake: overseas group + subject-group tree  
# Outputs used downstream:  
#   - probs_overseas_group      : P(overseas_group | overseas observed, excl Unknown)  
#   - probs_subject_by_group    : P(subject_group | overseas_group, excl Unknown)  
#   - probs_subject_overseas    : list(overseas_group -> named prob vector)  
# -----  
  
file_nat <- "table_10_candidates_subject_by_characteristics.csv"  
if (!file.exists(file_nat)) stop("Missing file: ", file_nat)  
  
tbl_nat <- readr::read_csv(file_nat, show_col_types = FALSE)  
  
# Column checks (fail fast if the file schema has changed)  
need_cols <- c("itt_subject", "nationality_group", "num_accepted", "sex", "age_grouping")  
missing_cols <- setdiff(need_cols, names(tbl_nat))  
if (length(missing_cols) > 0) {  
  stop("DfE file is missing expected columns: ", paste(missing_cols, collapse = ", "))  
}  
  
# Keep totals only to avoid double counting  
tbl_nat_clean <- tbl_nat %>%  
  dplyr::filter(  
    sex == "Total",  
    age_grouping == "Total",  
    ethnicity_major == "Total",  
    nationality_group != "Total"  
  ) %>%  
  dplyr::select(itt_subject, nationality_group, num_accepted)  
  
# Overseas tagging  
tbl_tagged <- tbl_nat_clean %>%
```

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dplyr::mutate(
  overseas_group = dplyr::case_when(
    nationality_group == "UK and Irish national" ~ "UK_IRL",
    nationality_group == "EEA national" ~ "EU_EEA",
    nationality_group == "Other nationality" ~ "OTHER_OVERSEAS",
    nationality_group == "Unknown" ~ "OVERSEAS_UNKNOWN",
    TRUE ~ "OVERSEAS_UNKNOWN"
  ),
  is_overseas_observed = overseas_group %in% c("EU_EEA", "OTHER_OVERSEAS", "OVERSEA
)

# Diagnostic: how much of "overseas observed" is Unknown (reportable)
unknown_check <- tbl_tagged %>%
  dplyr::filter(is_overseas_observed) %>%
  dplyr::summarise(
    overseas_total = sum(num_accepted, na.rm = TRUE),
    overseas_unknown = sum(num_accepted[overseas_group == "OVERSEAS_UNKNOWN"], na.rm
    pct_unknown = 100 * overseas_unknown / overseas_total
  )

# Model estimation excludes Unknown
tbl_overseas_model <- tbl_tagged %>%
  dplyr::filter(is_overseas_observed, overseas_group != "OVERSEAS_UNKNOWN") %>%
  dplyr::mutate(
    subject_group = dplyr::case_when(
      itt_subject == "Physics" ~ "Physics",
      itt_subject == "Modern Foreign Languages" ~ "Modern Languages",
      TRUE ~ "Other"
    )
  )

# P(overseas_group | overseas observed, excl Unknown)
probs_overseas_group <- tbl_overseas_model %>%
  dplyr::group_by(overseas_group) %>%
  dplyr::summarise(n = sum(num_accepted, na.rm = TRUE), .groups = "drop") %>%
  dplyr::mutate(p = n / sum(n)) %>%
  dplyr::select(overseas_group, p) %>%
  tibble::deframe()

assert_prob_vec(as.numeric(probs_overseas_group), "probs_overseas_group")

# P(subject_group | overseas_group), excluding Unknown
probs_subject_by_group <- tbl_overseas_model %>%
  dplyr::group_by(overseas_group, subject_group) %>%
  dplyr::summarise(n = sum(num_accepted, na.rm = TRUE), .groups = "drop") %>%
  dplyr::group_by(overseas_group) %>%
  dplyr::mutate(p = n / sum(n)) %>%
  dplyr::ungroup() %>%
  dplyr::select(overseas_group, subject_group, p) %>%

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dplyr::arrange(overseas_group, dplyr::desc(p))

# Convert to list of named vectors for simulation
probs_subject_overseas <- probs_subject_by_group %>%
  dplyr::group_split(overseas_group) %>%
  rlang::set_names(purrr::map_chr(., ~ unique(.x$overseas_group))) %>%
  purrr::map(~ .x %>% dplyr::select(subject_group, p) %>% tibble::deframe())

# Sanity checks on conditional probability vectors
cond_sums <- purrr::map_dbl(probs_subject_overseas, sum)
if (any(abs(cond_sums - 1) > 1e-8)) {
  stop("Some P(subject_group|overseas_group) vectors do not sum to 1: ",
       paste(names(cond_sums), signif(cond_sums, 8), collapse = "; "))
}

# Clean report tables
tbl_overseas_probs <- tibble::tibble(
  overseas_group = names(probs_overseas_group),
  p = as.numeric(probs_overseas_group)
) %>% dplyr::arrange(dplyr::desc(p))

tbl_subject_probs <- probs_subject_by_group %>%
  dplyr::mutate(p = as.numeric(p)) %>%
  dplyr::arrange(overseas_group, dplyr::desc(p))

tbl_subject_sums <- tbl_subject_probs %>%
  dplyr::group_by(overseas_group) %>%
  dplyr::summarise(sum_p = sum(p), .groups = "drop")

cat("✓ Block 1 complete: overseas_group + subject_group probability tree built\n")

```

```
## ✓ Block 1 complete: overseas_group + subject_group probability tree built
```

| overseas_total | overseas_unknown | pct_unknown |
|----------------|------------------|-------------|
| 38068          | 8                | 0.02        |

Overseas observed: share of 'Unknown' nationality\_group (excluded from modelled probabilities).

| overseas_group | p        |
|----------------|----------|
| OTHER_OVERSEAS | 0.620704 |
| EU_EEA         | 0.379296 |

Modelled overseas mix (excluding Unknown):  $P(\text{overseas\_group} \mid \text{overseas observed, excl Unknown})$ .

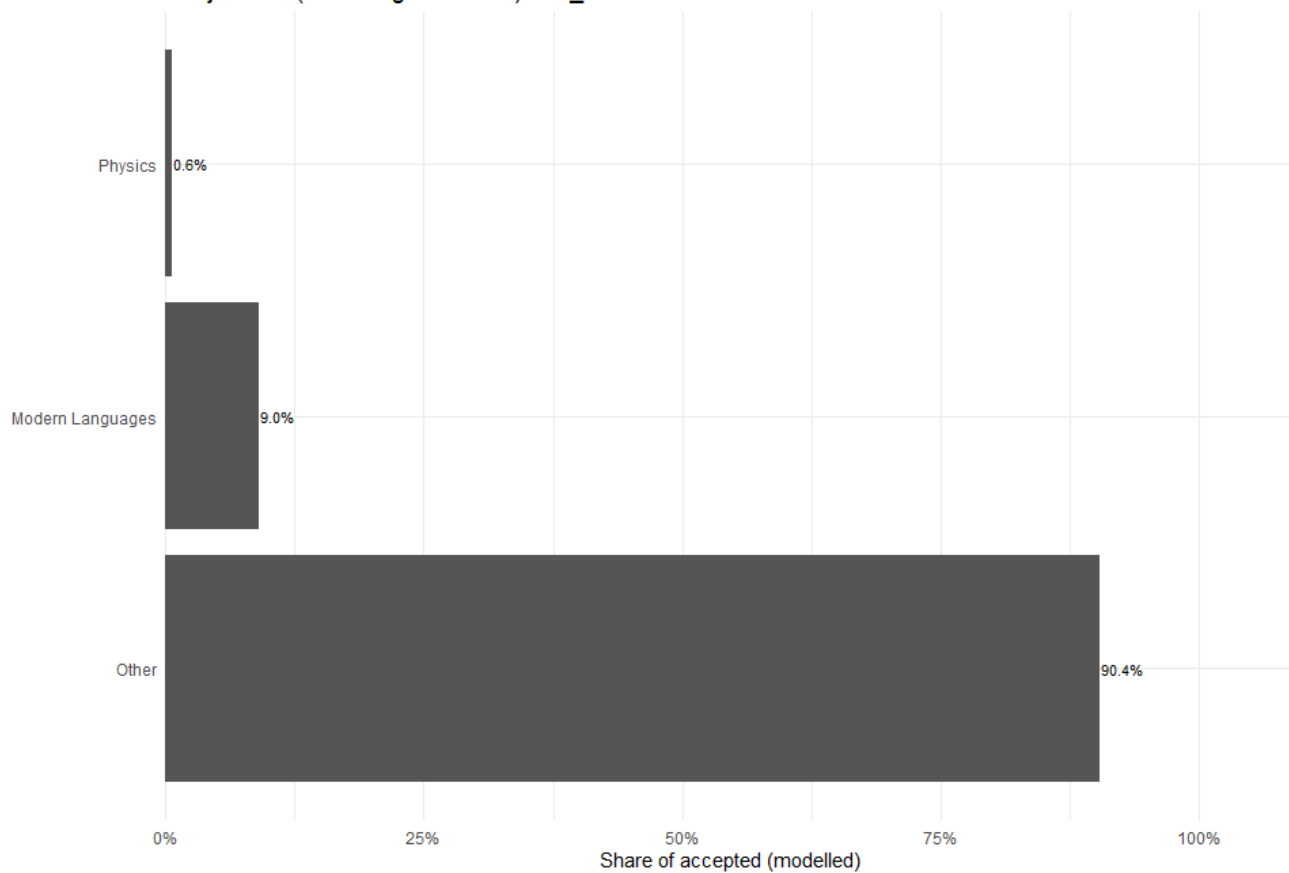
| <b>overseas_group</b> | <b>subject_group</b> | <b>p</b> |
|-----------------------|----------------------|----------|
| EU_EEA                | Other                | 0.904129 |
| EU_EEA                | Modern Languages     | 0.089845 |
| EU_EEA                | Physics              | 0.006027 |
| OTHER_OVERSEAS        | Other                | 0.903276 |
| OTHER_OVERSEAS        | Physics              | 0.069040 |
| OTHER_OVERSEAS        | Modern Languages     | 0.027684 |

Conditional subject mix (excluding Unknown):  $P(\text{subject\_group} \mid \text{overseas\_group})$ .

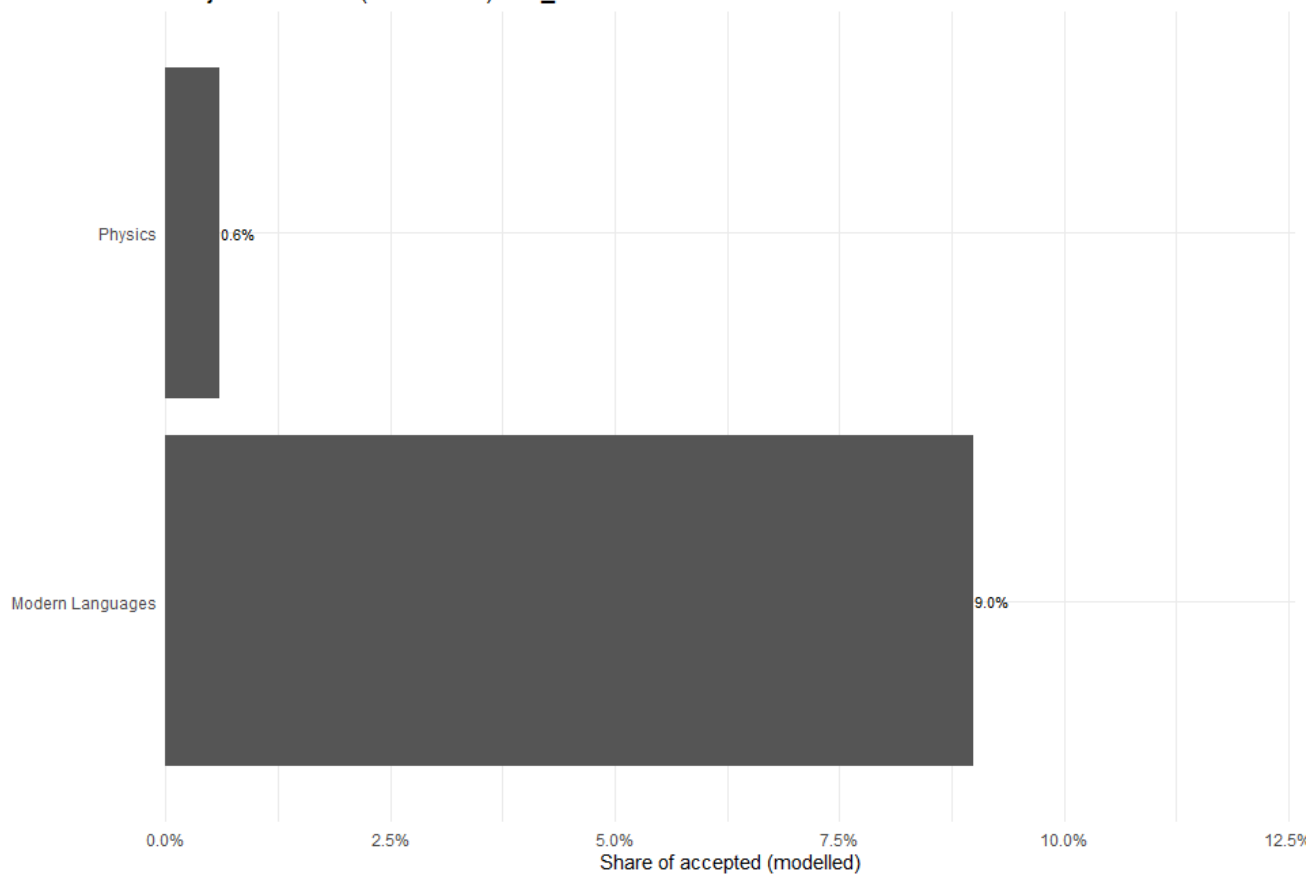
| <b>overseas_group</b> | <b>sum_p</b> |
|-----------------------|--------------|
| EU_EEA                | 1            |
| OTHER_OVERSEAS        | 1            |

Sanity check: sums of  $P(\text{subject\_group} \mid \text{overseas\_group})$  (should be 1).

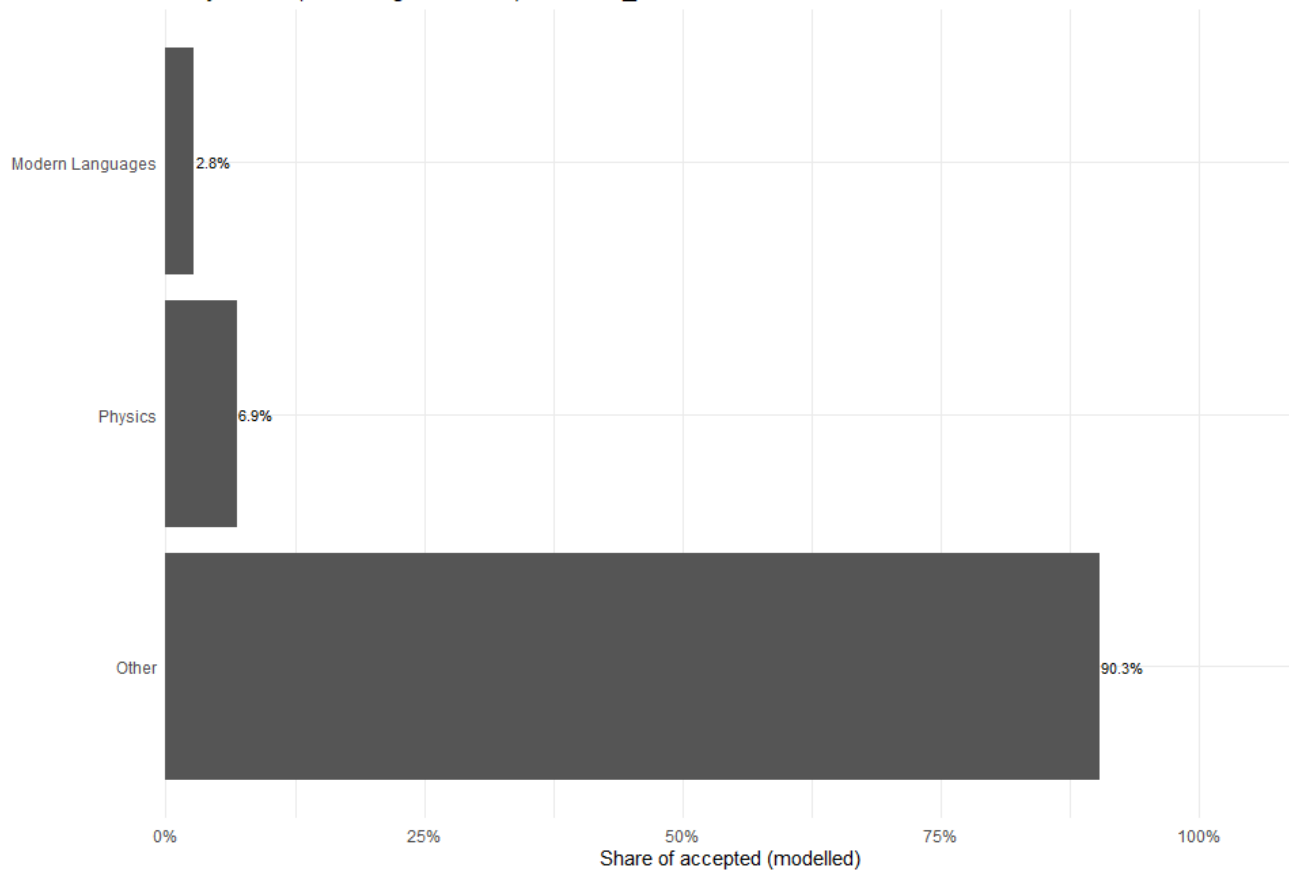
Subject mix (excluding Unknown): EU\_EEA



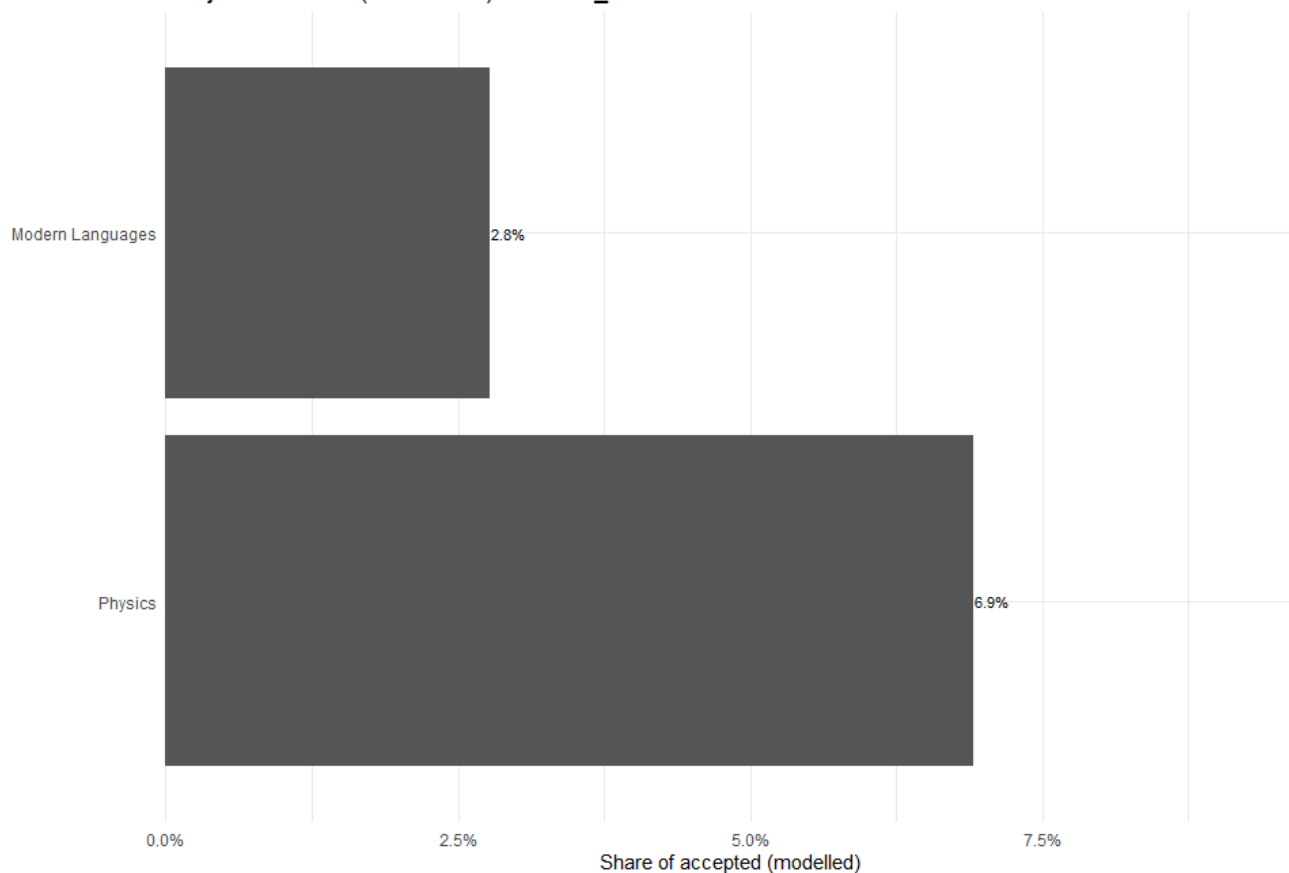
Subject mix zoom (excl 'Other'): EU\_EEA



Subject mix (excluding Unknown): OTHER\_OVERSEAS



Subject mix zoom (excl 'Other'): OTHER\_OVERSEAS



```

# -----
# Block 2 – Degree-class probabilities for overseas (pooled)
# Inputs from Block 1:
#   - probs_overseas_group
#   - probs_subject_overseas
# Output used downstream:
#   - probs_degree_overseas_all : P(degree_class | Overseas) pooled correctly
# -----

# Fallback (in case Block 0 wasn't run for some reason)
if (!exists("assert_prob_vec")) {
  assert_prob_vec <- function(p, name = "prob") {
    if (any(!is.finite(p))) stop(name, " has non-finite entries.")
    if (any(p < -1e-12)) stop(name, " has negative entries.")
    s <- sum(p)
    if (!is.finite(s) || abs(s - 1) > 1e-8) stop(name, " must sum to 1 (sum=", signif(
      invisible(TRUE)
    )
  )
}

file_deg <- "table4.csv"
if (!file.exists(file_deg)) stop("Missing file: ", file_deg)

tbl_degree <- readr::read_csv(file_deg, show_col_types = FALSE)

need_cols_deg <- c("time_period", "breakdown_topic", "breakdown", "itt_subject", "tra
missing_cols_deg <- setdiff(need_cols_deg, names(tbl_degree))
if (length(missing_cols_deg) > 0) {
  stop("Degree file is missing expected columns: ", paste(missing_cols_deg, collapse
}

degree_levels <- c("First", "Upper Second", "Lower Second", "Other")
subject_groups <- c("Physics", "Modern Languages", "Other")

# Counts by subject_group x degree_class (this is the key step)
deg_group_counts <- tbl_degree %>%
  dplyr::filter(
    time_period == 202526,
    breakdown_topic == "Degree Class",
    breakdown %in% degree_levels
  ) %>%
  dplyr::mutate(
    subject_group = dplyr::case_when(
      itt_subject == "Physics" ~ "Physics",
      itt_subject == "Modern Foreign Languages" ~ "Modern Languages",
      TRUE ~ "Other"
    )
  ) %>%
  dplyr::group_by(subject_group, breakdown) %>%

```

```

dplyr::summarise(n = sum(trainee_number, na.rm = TRUE), .groups = "drop")

# P(degree | subject_group)
p_deg_given_group <- deg_group_counts %>%
  dplyr::group_by(subject_group) %>%
  dplyr::mutate(p = n / sum(n)) %>%
  dplyr::ungroup() %>%
  dplyr::select(subject_group, degree_class = breakdown, p)

# Sanity: each conditional distribution sums to 1
deg_cond_sums <- p_deg_given_group %>%
  dplyr::group_by(subject_group) %>%
  dplyr::summarise(sum_p = sum(p), .groups = "drop")

if (any(abs(deg_cond_sums$sum_p - 1) > 1e-8)) {
  stop("Some P(degree|subject_group) sums are not 1:\n",
       paste(deg_cond_sums$subject_group, signif(deg_cond_sums$sum_p, 10), collapse =
  )
}

# Mix helper: given P(degree|subject_group) and a subject_group probability vector ->
mix_degree_probs <- function(p_deg_given_group, subject_group_probs,
                             subject_groups = c("Physics", "Modern Languages", "Other"),
                             degree_levels = c("First", "Upper Second", "Lower Second",

if (any(!subject_groups %in% names(subject_group_probs))) {
  miss <- setdiff(subject_groups, names(subject_group_probs))
  stop("Missing subject_group probabilities: ", paste(miss, collapse = ", "))
}

p_sub <- subject_group_probs[subject_groups]
p_sub <- as.numeric(p_sub)
names(p_sub) <- subject_groups
p_sub <- p_sub / sum(p_sub)
assert_prob_vec(p_sub, "subject_group_probs")

tmp <- p_deg_given_group %>%
  dplyr::filter(subject_group %in% subject_groups, degree_class %in% degree_levels)
  tidyr::complete(
    subject_group = subject_groups,
    degree_class = degree_levels,
    fill = list(p = 0)
  ) %>%
  dplyr::mutate(
    p_group = p_sub[subject_group],
    weighted = p * p_group
  )

out <- tmp %>%
  dplyr::group_by(degree_class) %>%

```



```

dplyr::summarise(p = sum(weighted), .groups = "drop") %>%
dplyr::mutate(p = p / sum(p))

vec <- out %>% dplyr::select(degree_class, p) %>% tibble::deframe()
vec <- vec[degree_levels]
assert_prob_vec(as.numeric(vec), "mixed degree probs")
vec
}

# Verification: compare EU_EEA vs OTHER_OVERSEAS degree mixes implied by their subject
deg_EU <- mix_degree_probs(p_deg_given_group, probs_subject_overseas$EU_EEA,
                           subject_groups = subject_groups, degree_levels = degree_le

deg_00 <- mix_degree_probs(p_deg_given_group, probs_subject_overseas$OTHER_OVERSEAS,
                           subject_groups = subject_groups, degree_levels = degree_le

verify_tbl <- tibble::tibble(
  degree_class = degree_levels,
  EU_EEA = as.numeric(deg_EU[degree_levels]),
  OTHER_OVERSEAS = as.numeric(deg_00[degree_levels]),
  abs_diff = abs(EU_EEA - OTHER_OVERSEAS)
)

verify_summary <- verify_tbl %>%
  dplyr::summarise(
    max_abs_diff = max(abs_diff),
    ll_distance = sum(abs_diff)
  )

# P(subject_group | Overseas) pooled via total probability:
# sum_g P(subject_group | g) P(g), where g in {EU_EEA, OTHER_OVERSEAS}
if (!all(names(probs_overseas_group) %in% names(probs_subject_overseas))) {
  stop("Mismatch: probs_overseas_group keys are not all present in probs_subject_over
}

probs_subject_overall_overseas <- setNames(
  vapply(subject_groups, function(sg) {
    sum(vapply(names(probs_overseas_group), function(g) {
      probs_subject_overseas[[g]][sg] * probs_overseas_group[g]
    }, numeric(1)))
  }, numeric(1)),
  subject_groups
)
probs_subject_overall_overseas <- probs_subject_overall_overseas / sum(probs_subject_
assert_prob_vec(as.numeric(probs_subject_overall_overseas), "probs_subject_overall_ov

# P(degree | Overseas) pooled (this is the distribution used in the simulator)
probs_degree_overseas_all <- mix_degree_probs(
  p_deg_given_group,

```

```

probs_subject_overall_overseas,
subject_groups = subject_groups,
degree_levels = degree_levels
)

cat("✓ Block 2 complete: pooled overseas degree probabilities ready\n")

```

```
## ✓ Block 2 complete: pooled overseas degree probabilities ready
```

```
print(probs_degree_overseas_all)
```

```
##           First Upper Second Lower Second           Other
##    0.2230514    0.4855497    0.1893497    0.1020492
```

| subject_group    | breakdown    | n     |
|------------------|--------------|-------|
| Modern Languages | Upper Second | 1098  |
| Modern Languages | First        | 530   |
| Modern Languages | Other        | 342   |
| Modern Languages | Lower Second | 262   |
| Other            | Upper Second | 58548 |
| Other            | First        | 26819 |
| Other            | Lower Second | 22703 |
| Other            | Other        | 10792 |
| Physics          | Upper Second | 562   |
| Physics          | Other        | 448   |
| Physics          | Lower Second | 396   |
| Physics          | First        | 258   |

Degree-class counts by subject group (DfE Table 4).

| <b>subject_group</b> | <b>degree_class</b> | <b>p</b> |
|----------------------|---------------------|----------|
| Modern Languages     | Upper Second        | 0.491935 |
| Modern Languages     | First               | 0.237455 |
| Modern Languages     | Other               | 0.153226 |
| Modern Languages     | Lower Second        | 0.117384 |
| Other                | Upper Second        | 0.492571 |
| Other                | First               | 0.225631 |
| Other                | Lower Second        | 0.191003 |
| Other                | Other               | 0.090794 |
| Physics              | Upper Second        | 0.337740 |
| Physics              | Other               | 0.269231 |
| Physics              | Lower Second        | 0.237981 |
| Physics              | First               | 0.155048 |

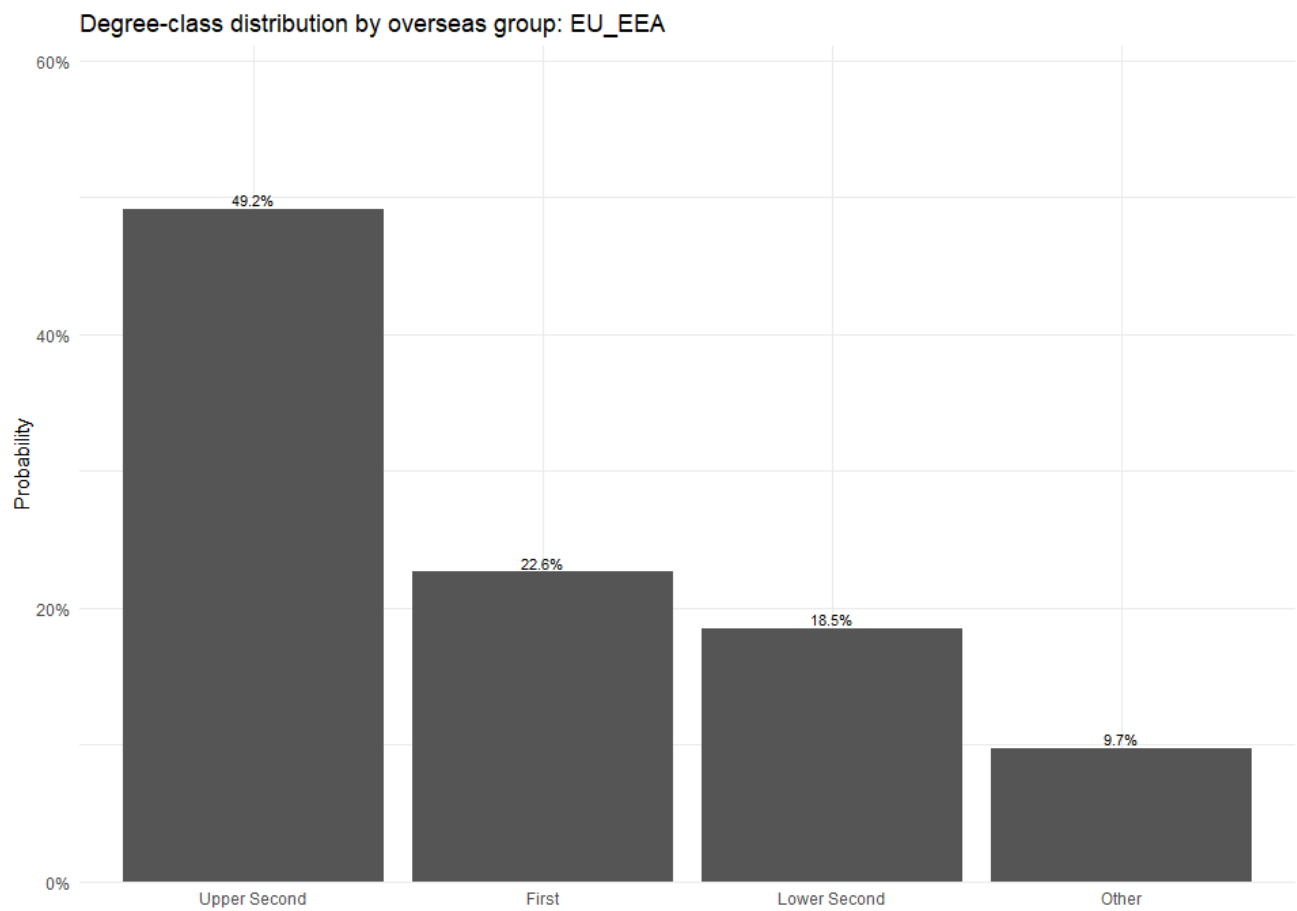
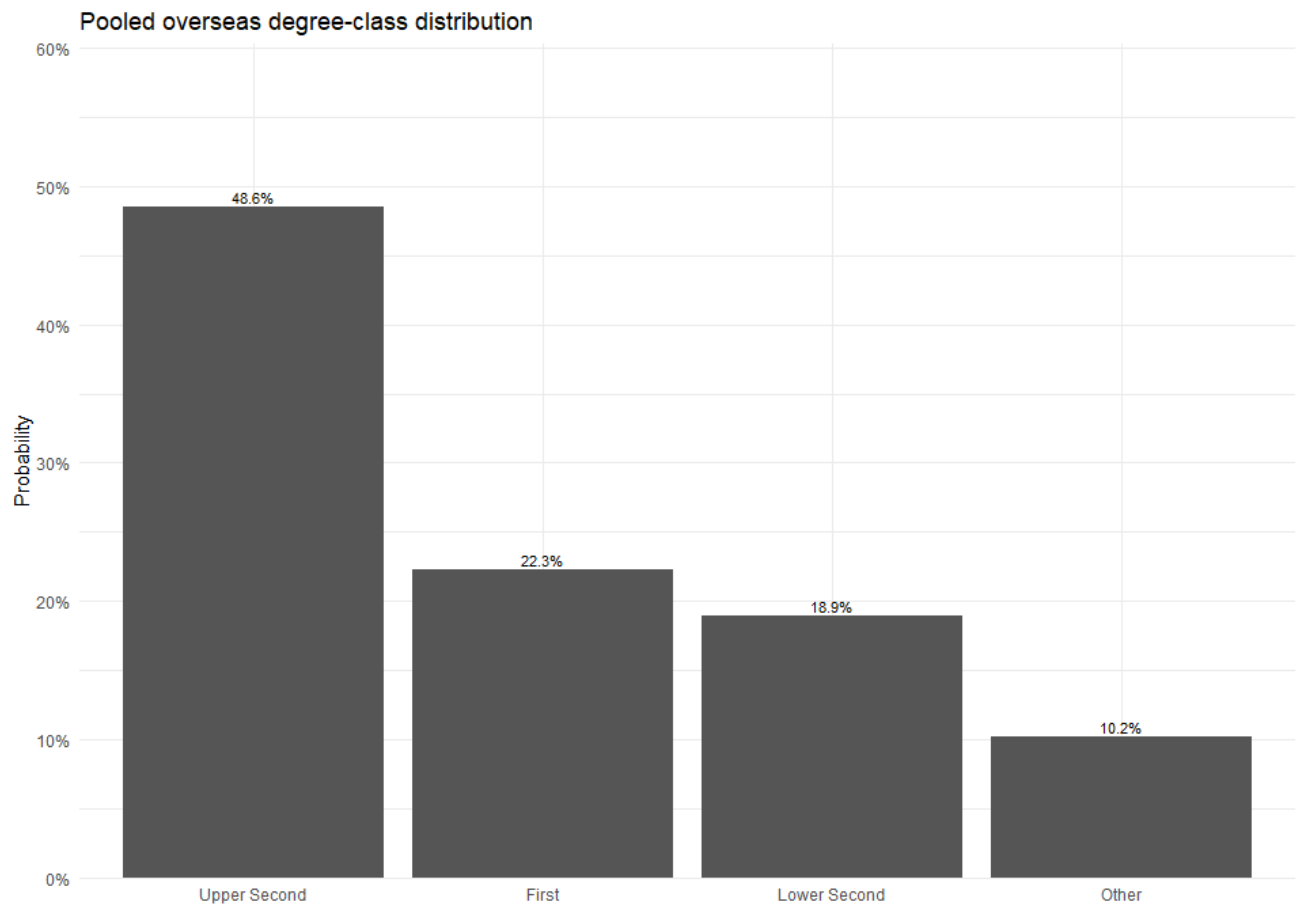
Conditional degree distribution:  $P(\text{degree\_class} \mid \text{subject\_group})$ .

| <b>degree_class</b> | <b>EU_EEA</b> | <b>OTHER_OVERSEAS</b> | <b>abs_diff</b> |
|---------------------|---------------|-----------------------|-----------------|
| First               | 0.226268      | 0.221086              | 0.005183        |
| Upper Second        | 0.491581      | 0.481864              | 0.009717        |
| Lower Second        | 0.184672      | 0.192208              | 0.007536        |
| Other               | 0.097479      | 0.104842              | 0.007363        |

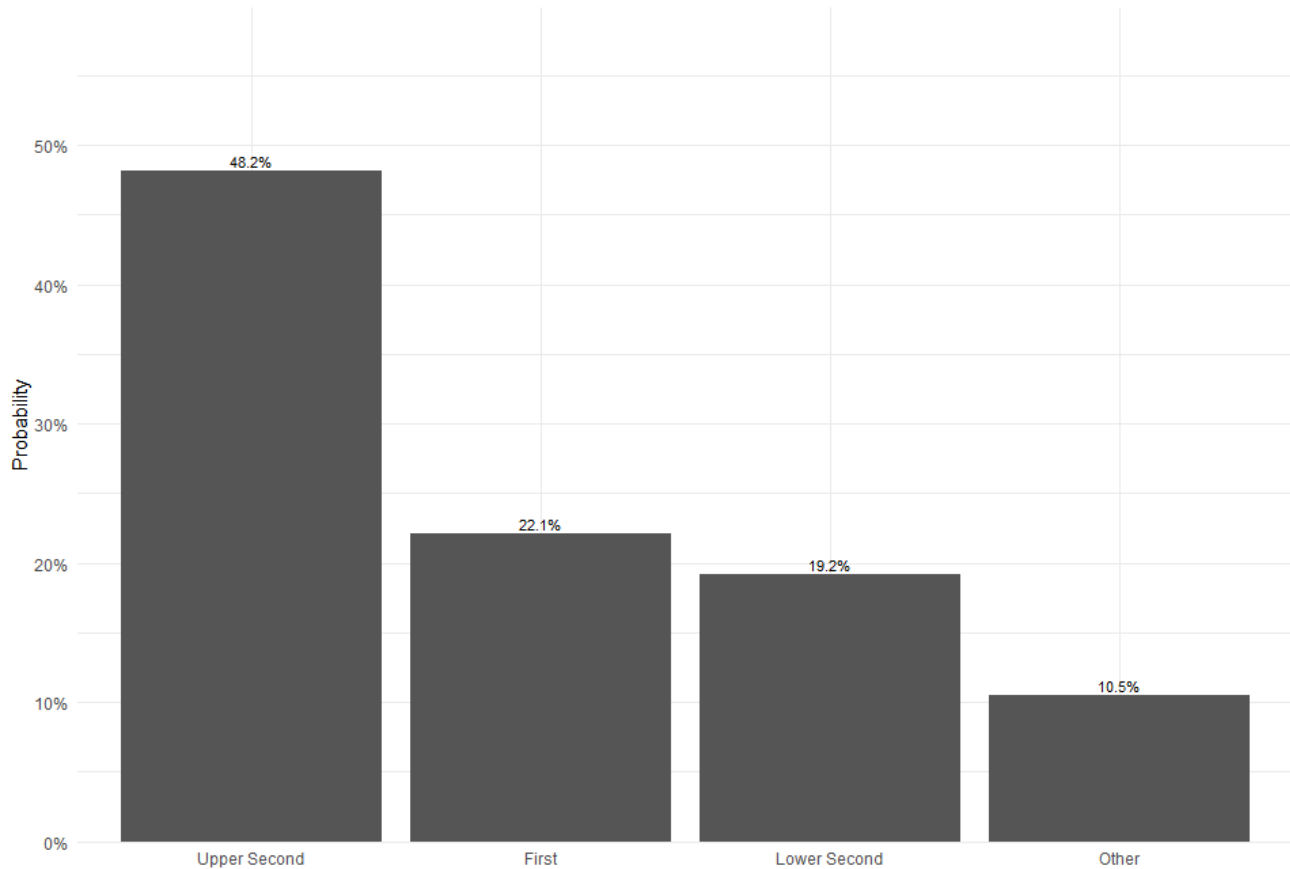
Verification: implied degree mix by overseas group (differences should be small).

| <b>max_abs_diff</b> | <b>l1_distance</b> |
|---------------------|--------------------|
| 0.009717            | 0.029799           |

Verification summary: maximum absolute difference and L1 distance.



Degree-class distribution by overseas group: OTHER\_OVERSEAS



```
# -----
# Block 3 – Policy layer + core simulation engine
#   Inputs from Block 1-2:
#     - probs_overseas_group
#     - probs_subject_overseas
#     - probs_degree_overseas_all
#   Outputs for Block 4:
#     - policy_A, policy_B, policy_C
#     - generate_potential_overseas(), add_training_and_offers(),
#       assign_training_incentives(), apply_policy_and_score()
# -----

# Minimal guardrails (in case Block 0 wasn't run)
if (!exists("assert_prob_vec")) {
  assert_prob_vec <- function(p, name = "prob") {
    if (any(!is.finite(p))) stop(name, " has non-finite entries.")
    if (any(p < -1e-12)) stop(name, " has negative entries.")
    s <- sum(p)
    if (!is.finite(s) || abs(s - 1) > 1e-8) stop(name, " must sum to 1 (sum=", signif(
invisible(TRUE)
  }
}

# -----
# 3.0 Core parameters
```

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# -----

# EU/EEA: probability they already have a status (no visa needed)
p_eu_has_euss <- 0.60

# Visa costs counted upfront for year-1 accounting
visa_years_upfront <- 3

# Narrative horizon (not charged upfront here)
years_to_settlement <- 5

# IHS and Skilled Worker fees (use your report's chosen numbers)
ihs_per_year <- 1035
visa_fee_sw_leq3 <- 769
visa_fee_sw_gt3 <- 1519

# School sponsorship costs (Certificate of Sponsorship + Immigration Skills Charge)
cos_fee <- 525
isc_per_year <- 364

# Fallback only (kept for interim compatibility; main engine will compute from policy
cost_school_sponsorship <- cos_fee + isc_per_year * visa_years_upfront

calc_school_sponsorship_cost <- function(policy, params) {
  yrs <- policy$visa_years_upfront
  if (is.null(yrs) || !is.finite(yrs)) {
    warning("Policy missing visa_years_upfront; falling back to global/default for sp
    yrs <- params$visa_years_upfront
  }
  params$cos_fee + params$isc_per_year * yrs
}

# Training incentives (toggle if you want the non-UK line for MFL)
use_nonuk_training_support <- TRUE

bursary_physics <- 29000
scholarship_physics <- 31000

# -----
# Modern Languages incentives – bursary-only (disable scholarships)
# Rationale: official language scholarships are restricted (e.g., F/G/S only);
# if the model does not split Modern Languages into eligible subtypes,
# we disable scholarships to avoid over-claiming.
# -----

if (use_nonuk_training_support) {
  bursary_languages <- 20000
  scholarship_languages <- 22000 # kept for reference; NOT USED when prob = 0
} else {

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bursary_languages      <- 26000
scholarship_languages <- 28000 # kept for reference; NOT USED when prob = 0
}

prob_scholarship_physics <- 0.15
# Hard-disable Modern Languages scholarships in the awarding mechanism
prob_scholarship_languages <- 0

# Relocation support (DfE)
relocation_support_amount <- 10000
relocation_subjects <- c("Physics", "Modern Languages") # set to NULL to apply to all

# Capacity constraint (posts available)
capacity_posts <- 1200

# Integration course (school cost, visa-friction recruits only)
integration_course_cost_per_trainee <- 400

# "Exceptional" marker for scholarship eligibility proxy
prob_exceptional <- 0.15

# Retention baselines
retention_baseline_no_visa      <- 0.78
retention_baseline_visa_noR     <- 0.60
retention_baseline_visa_withR   <- 0.75
integration_boost_visa <- 0.06

# Consolidate core constants (used to remove globals inside functions later)
params <- list(
  p_eu_has_euss = p_eu_has_euss,
  visa_years_upfront = visa_years_upfront,
  years_to_settlement = years_to_settlement,
  ihs_per_year = ihs_per_year,
  visa_fee_sw_leq3 = visa_fee_sw_leq3,
  visa_fee_sw_gt3 = visa_fee_sw_gt3,
  cos_fee = cos_fee,
  isc_per_year = isc_per_year,
  use_nonuk_training_support = use_nonuk_training_support,
  bursary_physics = bursary_physics,
  scholarship_physics = scholarship_physics,
  bursary_languages = bursary_languages,
  scholarship_languages = scholarship_languages,
  prob_scholarship_physics = prob_scholarship_physics,
  prob_scholarship_languages = prob_scholarship_languages,
  relocation_support_amount = relocation_support_amount,
  relocation_subjects = relocation_subjects,
  capacity_posts = capacity_posts,
  integration_course_cost_per_trainee = integration_course_cost_per_trainee,
  prob_exceptional = prob_exceptional,

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    retention_baseline_no_visa = retention_baseline_no_visa,
    retention_baseline_visa_noR = retention_baseline_visa_noR,
    retention_baseline_visa_withR = retention_baseline_visa_withR,
    integration_boost_visa = integration_boost_visa
)
params$prob_scholarship_languages <- 0

# -----
# 3.1 Policy scenarios (A/B/C)
# -----

policy_A <- list(
  name = "Status Quo (S=0.40, R=0, I=0)",
  prob_sponsor = 0.40,
  visa_years_upfront = visa_years_upfront,
  years_to_settlement = years_to_settlement,
  ihs_per_year = ihs_per_year,
  visa_fee_sw_leq3 = visa_fee_sw_leq3,
  visa_fee_sw_gt3 = visa_fee_sw_gt3,
  relocation_support = 0,
  retention_baseline_no_visa = retention_baseline_no_visa,
  retention_baseline_visa = retention_baseline_visa_noR,
  integration_coverage = 0.00,
  integration_boost_visa = 0.00
)

policy_B <- list(
  name = "Auto Sponsor + Relocation (S=1.00, R=1, I=0)",
  prob_sponsor = 1.00,
  visa_years_upfront = visa_years_upfront,
  years_to_settlement = years_to_settlement,
  ihs_per_year = ihs_per_year,
  visa_fee_sw_leq3 = visa_fee_sw_leq3,
  visa_fee_sw_gt3 = visa_fee_sw_gt3,
  relocation_support = relocation_support_amount,
  retention_baseline_no_visa = retention_baseline_no_visa,
  retention_baseline_visa = retention_baseline_visa_withR,
  integration_coverage = 0.00,
  integration_boost_visa = 0.00
)

policy_C <- list(
  name = "Auto Sponsor + Relocation + Integration (S=1.00, R=1, I=1)",
  prob_sponsor = 1.00,
  visa_years_upfront = visa_years_upfront,
  years_to_settlement = years_to_settlement,
  ihs_per_year = ihs_per_year,
  visa_fee_sw_leq3 = visa_fee_sw_leq3,

```



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visa_fee_sw_gt3 = visa_fee_sw_gt3,
relocation_support = relocation_support_amount,
retention_baseline_no_visa = retention_baseline_no_visa,
retention_baseline_visa = retention_baseline_visa_withR,
integration_coverage = 1.00,
integration_boost_visa = integration_boost_visa
)

cat("✓ Block 3 parameters + policies defined\n")

```

```
## ✓ Block 3 parameters + policies defined
```

## Accounting conventions (spend vs waste; year-1 horizon)

**Horizon.** All accounting is reported on a **year-1 horizon**. “Retained (year 1)” means recruited and still in post at the end of year 1. Any trainee/teacher who leaves by end of year 1 is treated as a year-1 loss.

**DfE (public) spend.** - **Spend** includes: training incentives awarded at training stage (**bursary/scholarship**) plus any **relocation support** paid to eligible recruited trainees. -

**Waste** includes: - bursary/scholarship paid to trainees who **do not become recruited** or who **leave by end of year 1**; and - relocation support paid to recruits who **leave by end of year 1**. - Rationale: bursaries/scholarships are policy spend intended to increase the supply of retained teachers; if that conversion fails within the year-1 window, the spend is counted as wasted for this horizon.

**School (provider/employer) spend.** - **Spend** includes: sponsorship cost (CoS + ISC) for recruited visa-exposed teachers, plus any integration course cost for those assigned the course. - **Waste** includes school spend attached to recruited teachers who **leave by end of year 1**.

**Teacher (private) spend.** - **Spend** includes: upfront visa/IHS costs borne by the teacher **net of any relocation support** (i.e., out-of-pocket after relocation offsets). - **Waste** includes teacher net spend attached to recruited teachers who **leave by end of year 1**.

```

# -----
# 3.2 Simulation helpers
# -----

```

```

inv_logit <- function(x) 1 / (1 + exp(-x))

sample_cat <- function(n, prob_vec) {
  assert_prob_vec(as.numeric(prob_vec), "prob_vec")
  sample(names(prob_vec), n, replace = TRUE, prob = prob_vec)
}

# 3.2.1 Generate potential overseas candidates (accepted to ITT)
generate_potential_overseas <- function(
  n,
  probs_overseas_group,
  probs_subject_overseas,
  probs_degree_overseas_all,
  p_eu_has_euss,
  prob_exceptional
) {
  assert_prob_vec(as.numeric(probs_overseas_group), "probs_overseas_group")
  assert_prob_vec(as.numeric(probs_degree_overseas_all), "probs_degree_overseas_all")
  if (!all(c("EU_EEA", "OTHER_OVERSEAS") %in% names(probs_subject_overseas))) {
    stop("probs_subject_overseas must include EU_EEA and OTHER_OVERSEAS.")
  }

  overseas_group <- sample_cat(n, probs_overseas_group)

  needs_visa <- dplyr::case_when(
    overseas_group == "EU_EEA" ~ (runif(n) > p_eu_has_euss),
    overseas_group == "OTHER_OVERSEAS" ~ TRUE,
    TRUE ~ TRUE
  )

  subject <- character(n)
  idx_eu <- which(overseas_group == "EU_EEA")
  idx_oo <- which(overseas_group == "OTHER_OVERSEAS")

  if (length(idx_eu) > 0) subject[idx_eu] <- sample_cat(length(idx_eu), probs_subject)
  if (length(idx_oo) > 0) subject[idx_oo] <- sample_cat(length(idx_oo), probs_subject)

  degree_class <- sample_cat(n, probs_degree_overseas_all)

  tibble::tibble(
    id = seq_len(n),
    overseas_group = overseas_group,
    needs_visa = needs_visa,
    subject = subject,
    degree_class = degree_class,
    has_exceptional_qual = (rbinom(n, 1, prob_exceptional) == 1),
    induction_quality = runif(n)
  )
}

```

```

# 3.2.2 Training completion/QTS gate + capacity-limited offers
add_training_and_offers <- function(
  df_potential,
  capacity_posts,
  shortage_subjects = c("Physics", "Modern Languages"),
  u_qual = NULL,
  seed_offer = NULL
) {
  df <- df_potential %>%
    dplyr::mutate(
      degree_simple = dplyr::case_when(
        degree_class == "First" ~ "First",
        degree_class == "Upper Second" ~ "Upper Second",
        degree_class == "Lower Second" ~ "Lower Second",
        TRUE ~ "Other"
      )
    )

  n <- nrow(df)
  if (is.null(u_qual)) u_qual <- runif(n)
  if (length(u_qual) != n) stop("u_qual must have length nrow(df_potential).")

  beta_degree <- c("First"=0.55, "Upper Second"=0.30, "Lower Second"=0.10, "Other"=0.
  deg_eff <- unname(beta_degree[df$degree_simple]); deg_eff[is.na(deg_eff)] <- 0

  p_qualified <- inv_logit(-0.25 + 1.10*df$induction_quality + 0.50*as.numeric(df$has
  qualified <- (u_qual < p_qualified)

  offer_bonus <- c("First"=0.35, "Upper Second"=0.20, "Lower Second"=0.05, "Other"=0.
  deg_bonus <- unname(offer_bonus[df$degree_simple]); deg_bonus[is.na(deg_bonus)] <-
  offer_weight <- 1.0 + if_else(df$subject %in% shortage_subjects, 0.8, 0) + deg_bonu

  offer <- rep(FALSE, n)
  idx <- which(qualified)
  K <- min(capacity_posts, length(idx))

  restore_seed <- NULL
  if (!is.null(seed_offer)) {
    if (exists(".Random.seed", envir = .GlobalEnv, inherits = FALSE)) {
      restore_seed <- get(".Random.seed", envir = .GlobalEnv)
    }
    set.seed(seed_offer)
  }

  if (K > 0) {
    winners <- sample(idx, size = K, replace = FALSE, prob = pmax(offer_weight[idx],
    offer[winners] <- TRUE
  }

```

```

if (!is.null(seed_offer)) {
  if (is.null(restore_seed)) {
    if (exists(".Random.seed", envir = .GlobalEnv, inherits = FALSE)) rm(".Random.s
  } else {
    assign(".Random.seed", restore_seed, envir = .GlobalEnv)
  }
}

df %>%
  dplyr::mutate(
    p_qualified = p_qualified,
    qualified = qualified,
    offer_weight = offer_weight,
    offered = offer
  )
}

# 3.2.3 Assign bursary/scholarship at training stage
assign_training_incentives <- function(df_in, params = params) {
  if (is.null(params)) stop("assign_training_incentives(): params must be provided.")

  df <- df_in %>%
    dplyr::mutate(
      bursary_eligible = subject %in% c("Physics", "Modern Languages") &
        degree_simple %in% c("First", "Upper Second", "Lower Second"),

      scholarship_eligible = subject %in% c("Physics", "Modern Languages") &
        (degree_simple %in% c("First", "Upper Second") |
          (degree_simple == "Lower Second" & has_exceptional_qual)),

      scholarship_weight = dplyr::case_when(
        degree_simple == "First" ~ 3,
        degree_simple == "Upper Second" ~ 2,
        degree_simple == "Lower Second" ~ 1,
        TRUE ~ 0
      ) + 0.001
    )

  award_weighted_binom <- function(df0, subj, p_award) {
    idx <- which(df0$subject == subj & df0$scholarship_eligible)
    out <- rep(FALSE, nrow(df0))
    if (length(idx) == 0) return(out)

    K <- rbinom(1, size = length(idx), prob = p_award)
    if (K <= 0) return(out)

    winners <- sample(idx, size = min(K, length(idx)), replace = FALSE, prob = df0$sc
    out[winners] <- TRUE
  }
}

```

```

    out
  }

  sch_award <- rep(FALSE, nrow(df))
  sch_award <- sch_award | award_weighted_binom(df, "Physics", params$prob_scholarshi
  sch_award <- sch_award | award_weighted_binom(df, "Modern Languages", params$prob_s

  df %>%
    dplyr::mutate(
      scholarship_awarded = sch_award,
      incentive_type = dplyr::case_when(
        scholarship_awarded ~ "Scholarship",
        bursary_eligible ~ "Bursary",
        TRUE ~ "None"
      ),
      incentive_value = dplyr::case_when(
        subject == "Physics" & incentive_type == "Scholarship" ~ params$scholarship_p
        subject == "Physics" & incentive_type == "Bursary" ~ params$bursary_physics,
        subject == "Modern Languages" & incentive_type == "Scholarship" ~ params$scho
        subject == "Modern Languages" & incentive_type == "Bursary" ~ params$bursary_
        TRUE ~ 0
      )
    )
  }

# 3.2.4 Apply policy after offers exist + year-1 retention + spend/waste accounting
apply_policy_and_score <- function(
  df_pipeline,
  policy,
  cost_school_sponsorship = NULL, # legacy arg kept for compatibility; policy-derive
  relocation_subjects = c("Physics", "Modern Languages"),
  u_integration = NULL,
  u_sponsor = NULL,
  u_retention = NULL,
  params = params
) {
  if (is.null(params)) stop("apply_policy_and_score(): params must be provided.")

  df <- df_pipeline
  n <- nrow(df)

  if (is.null(u_integration)) u_integration <- runif(n)
  if (is.null(u_sponsor)) u_sponsor <- runif(n)
  if (is.null(u_retention)) u_retention <- runif(n)

  if (length(u_integration) != n) stop("u_integration length mismatch.")
  if (length(u_sponsor) != n) stop("u_sponsor length mismatch.")
  if (length(u_retention) != n) stop("u_retention length mismatch.")

```

```

sponsored <- df$offered & df$needs_visa & (u_sponsor < policy$prob_sponsor)
recruited <- df$offered & (!df$needs_visa | sponsored)

visa_fee_sw <- if (policy$visa_years_upfront <= 3) policy$visa_fee_sw_leq3 else pol
upfront_cost_gross <- if_else(
  recruited & df$needs_visa,
  visa_fee_sw + policy$ihs_per_year * policy$visa_years_upfront,
  0
)

relocation_eligible <- if (is.null(relocation_subjects)) {
  recruited & df$qualified & df$needs_visa
} else {
  recruited & df$qualified & df$needs_visa & (df$subject %in% relocation_subjects)
}
relocation_paid <- if_else(relocation_eligible, policy$relocation_support, 0)
upfront_cost_net_teacher <- pmax(0, upfront_cost_gross - relocation_paid)

integration_course <- recruited & df$needs_visa & (u_integration < policy$integrati
school_integration_spend <- params$integration_course_cost_per_trainee * as.numeric

# Policy-dependent sponsorship unit cost
sponsor_unit_cost <- calc_school_sponsorship_cost(policy, params)
school_sponsorship_spend <- sponsor_unit_cost * as.numeric(recruited & df$needs_vis

base_retention <- dplyr::case_when(
  !recruited ~ NA_real_,
  recruited & !df$needs_visa ~ pmin(1, policy$retention_baseline_no_visa + 0.20 * d
  recruited & df$needs_visa ~ pmin(1, policy$retention_baseline_visa + 0.35 * d
)
integration_boost <- if_else(recruited & df$needs_visa & integration_course, policy
retention_prob <- if_else(!recruited, NA_real_, pmin(1, base_retention + integratio

retained_year_1 <- if_else(recruited, (u_retention < retention_prob), NA)
left_year_1 <- recruited & !retained_year_1

# Spend (year-1 horizon)
dfe_spend <- df$incentive_value + relocation_paid
school_spend <- school_sponsorship_spend + school_integration_spend
teacher_spend <- upfront_cost_net_teacher

# Waste components (year-1)
dfe_incentive_waste <- if_else(df$incentive_value > 0 & (!recruited | left_year_1),
dfe_relocation_waste <- if_else(left_year_1, relocation_paid, 0)

school_sponsorship_waste <- if_else(left_year_1 & recruited & df$needs_visa, sponso
school_integration_waste <- if_else(left_year_1, school_integration_spend, 0)

teacher_money_wasted <- if_else(left_year_1, upfront_cost_net_teacher, 0)

```

```

df_out <- df %>%
  dplyr::mutate(
    sponsored = sponsored,
    recruited = recruited,
    retention_prob = retention_prob,
    retained_year_1 = retained_year_1,
    left_year_1 = left_year_1,
    relocation_eligible = relocation_eligible,
    relocation_paid = relocation_paid,
    upfront_cost_gross = upfront_cost_gross,
    upfront_cost_net_teacher = upfront_cost_net_teacher,
    integration_course = integration_course,
    school_integration_spend = school_integration_spend,
    school_sponsorship_spend = school_sponsorship_spend,
    dfe_spend = dfe_spend,
    school_spend = school_spend,
    teacher_spend = teacher_spend,
    dfe_waste = dfe_incentive_waste + dfe_relocation_waste,
    school_waste = school_sponsorship_waste + school_integration_waste,
    teacher_waste = teacher_money_wasted
  )

summary <- df_out %>%
  dplyr::summarise(
    policy = policy$name,
    n_potential = n(),
    n_qualified = sum(qualified),
    n_offered = sum(offered),
    n_recruited = sum(recruited),
    n_sponsored = sum(sponsored),

    # recruitment_rate = recruited share (of simulated accepted cohort)
    recruitment_rate = mean(recruited),

    attrition_rate_year1 = if_else(sum(recruited) > 0, mean(!retained_year_1[recruited]), NA_real_),
    share_needs_visa_among_recruited = if_else(sum(recruited) > 0, mean(needs_visa[recruited]), NA_real_),
    share_integration_among_recruited_needs_visa = if_else(
      sum(recruited & needs_visa) > 0,
      mean(integration_course[recruited & needs_visa]),
      NA_real_
    ),

    total_dfe_spend = sum(dfe_spend),
    total_dfe_waste = sum(dfe_waste),
    total_dfe_waste_visa_only = sum(dfe_waste[needs_visa], na.rm = TRUE),
    total_dfe_waste_nonvisa = sum(dfe_waste[!needs_visa], na.rm = TRUE),

    total_school_spend = sum(school_spend),

```

```

    total_school_waste = sum(school_waste),

    total_teacher_spend = sum(teacher_spend),
    total_teacher_waste = sum(teacher_waste),

    total_relocation_spend = sum(relocation_paid),
    relocation_spend_wasted = sum(if_else(left_year_1, relocation_paid, 0)),
    .groups = "drop"
  ) %>%
  dplyr::mutate(
    dfe_waste_rate = if_else(total_dfe_spend > 0, total_dfe_waste / total_dfe_spend,
    school_waste_rate = if_else(total_school_spend > 0, total_school_waste / total_school_spend,
    teacher_waste_rate = if_else(total_teacher_spend > 0, total_teacher_waste / total_teacher_spend,

    recruited_over_capacity = n_recruited / params$capacity_posts,
    recruited_over_qualified = if_else(n_qualified > 0, n_recruited / n_qualified,
  )

  list(df = df_out, summary = summary)
}

cat("✓ Block 3 functions loaded\n")

```

```
## ✓ Block 3 functions loaded
```

```
## ✓ Fast inner-loop functions sourced
```

```

# -----
# 3.3 Single run sanity checks + clean summary tables
# -----

set.seed(2025)
N0 <- 2000

potential <- generate_potential_overseas(
  n = N0,
  probs_overseas_group = probs_overseas_group,
  probs_subject_overseas = probs_subject_overseas,
  probs_degree_overseas_all = probs_degree_overseas_all,
  p_eu_has_euss = params$p_eu_has_euss,
  prob_exceptional = params$prob_exceptional
)

# Sanity: implied vs simulated visa-friction share
implied_needs_visa <- as.numeric(probs_overseas_group["OTHER_OVERSEAS"]) +
  as.numeric(probs_overseas_group["EU_EEA"]) * (1 - params$p_eu_has_euss)

```



```

sanity_visa <- tibble::tibble(
  implied_needs_visa = implied_needs_visa,
  simulated_needs_visa = mean(potential$needs_visa)
)

pipeline <- potential %>%
  add_training_and_offers(
    df_potential = .,
    capacity_posts = params$capacity_posts,
    u_qual = runif(nrow(potential)),
    seed_offer = 777
  ) %>%
  assign_training_incentives(params = params)

# Common random numbers (fixed across policies for this pipeline)
u_sponsor_all <- runif(nrow(pipeline))
u_retention_all <- runif(nrow(pipeline))
u_integration_all <- runif(nrow(pipeline))

# A/B/C (same pipeline + same CRN)
res_A <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_A,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)
res_B <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_B,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)
res_C <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_C,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)

```

```

abc_tbl <- dplyr::bind_rows(res_A$summary, res_B$summary, res_C$summary) %>%
  dplyr::mutate(
    total_system_waste = total_dfe_waste + total_school_waste + total_teacher_waste,
    retained_year1 = n_recruited * (1 - attrition_rate_year1),
    waste_per_retained = total_system_waste / pmax(retained_year1, 1)
  )

# 2x2 for sponsorship regimes (same pipeline + same CRN)
retention_visa_noR <- policy_A$retention_baseline_visa
retention_visa_withR <- policy_B$retention_baseline_visa
relocation_amount <- policy_B$relocation_support
integration_boost <- policy_C$integration_boost_visa

make_2x2 <- function(prob_sponsor_fixed, label_prefix) {
  base <- modifyList(policy_A, list(
    name = paste0(label_prefix, " None (R=0, I=0)"),
    prob_sponsor = prob_sponsor_fixed,
    relocation_support = 0,
    retention_baseline_visa = retention_visa_noR,
    integration_coverage = 0,
    integration_boost_visa = 0
  ))
  R_only <- modifyList(base, list(
    name = paste0(label_prefix, " Relocation only (R=1, I=0)"),
    relocation_support = relocation_amount,
    retention_baseline_visa = retention_visa_withR
  ))
  I_only <- modifyList(base, list(
    name = paste0(label_prefix, " Integration only (R=0, I=1)"),
    integration_coverage = 1.0,
    integration_boost_visa = integration_boost
  ))
  R_I <- modifyList(base, list(
    name = paste0(label_prefix, " Relocation + Integration (R=1, I=1)"),
    relocation_support = relocation_amount,
    retention_baseline_visa = retention_visa_withR,
    integration_coverage = 1.0,
    integration_boost_visa = integration_boost
  ))
  list(base, R_only, I_only, R_I)
}

run_policy_list <- function(policy_list) {
  purrr::map_dfr(policy_list, function(pol) {
    apply_policy_and_score(
      df_pipeline = pipeline,
      policy = pol,
      relocation_subjects = params$relocation_subjects,
      u_integration = u_integration_all,

```

```

    u_sponsor = u_sponsor_all,
    u_retention = u_retention_all,
    params = params
  )$summary
}) %>%
  dplyr::mutate(
    total_system_waste = total_dfe_waste + total_school_waste + total_teacher_waste
  )
}

tbl_2x2_SQ <- run_policy_list(make_2x2(0.40, "S=0.40 |"))
tbl_2x2_AS <- run_policy_list(make_2x2(1.00, "S=1.00 |"))

cat("✓ Block 3 single-run objects ready: sanity_visa, abc_tbl, tbl_2x2_SQ, tbl_2x2_AS")

```

```
## ✓ Block 3 single-run objects ready: sanity_visa, abc_tbl, tbl_2x2_SQ, tbl_2x2_AS
```

| implied_needs_visa | simulated_needs_visa |
|--------------------|----------------------|
| 0.7724             | 0.757                |

Sanity check: implied vs simulated visa-friction share (single run).

| Policy                                                   | Recruited | Recruited<br>share (of<br>simulated<br>accepted<br>cohort) | Recruited<br>/ capacity | Recruited<br>/ qualified | Attrition<br>(yr1) | Ne<br>visa s<br>(recru |
|----------------------------------------------------------|-----------|------------------------------------------------------------|-------------------------|--------------------------|--------------------|------------------------|
| Status<br>Quo<br>(S=0.40,<br>R=0, I=0)                   | 647       | 0.3235                                                     | 0.5392                  | 0.4854                   | 0.1515             | 0.                     |
| Auto<br>Sponsor +<br>Relocation<br>(S=1.00,<br>R=1, I=0) | 1200      | 0.6000                                                     | 1.0000                  | 0.9002                   | 0.0742             | 0.                     |
| Auto<br>Sponsor +                                        | 1200      | 0.6000                                                     | 1.0000                  | 0.9002                   | 0.0483             | 0.                     |

| <b>Policy</b>                                           | <b>Recruited</b> | <b>Recruited<br/>share (of<br/>simulated<br/>accepted<br/>cohort)</b> | <b>Recruited<br/>/ capacity</b> | <b>Recruited<br/>/ qualified</b> | <b>Attrition<br/>(yr1)</b> | <b>Net<br/>visa s<br/>(recru</b> |
|---------------------------------------------------------|------------------|-----------------------------------------------------------------------|---------------------------------|----------------------------------|----------------------------|----------------------------------|
| Relocation<br>+<br>Integration<br>(S=1.00,<br>R=1, I=1) |                  |                                                                       |                                 |                                  |                            |                                  |

Policies A/B/C outcomes (single run, common random numbers).

| <b>Policy</b>                                                                | <b>DfE spend</b> | <b>DfE waste</b> | <b>DfE<br/>waste<br/>rate</b> | <b>School<br/>spend</b> | <b>School<br/>waste</b> | <b>School<br/>waste<br/>rate</b> |
|------------------------------------------------------------------------------|------------------|------------------|-------------------------------|-------------------------|-------------------------|----------------------------------|
| Status<br>Quo<br>(S=0.40,<br>R=0, I=0)                                       | £3,918,000       | £2,980,000       | 76.1%                         | £540,078                | £114,807                | 21.3%                            |
| Auto<br>Sponsor +<br>Relocation<br>(S=1.00,<br>R=1, I=0)                     | £4,858,000       | £1,675,000       | 34.5%                         | £1,434,279              | £100,254                | 7.0%                             |
| Auto<br>Sponsor +<br>Relocation<br>+<br>Integration<br>(S=1.00,<br>R=1, I=1) | £4,858,000       | £1,564,000       | 32.2%                         | £1,789,079              | £62,527                 | 3.5%                             |

Policies A/B/C accounting (year-1 horizon; spend, waste, and waste rates).

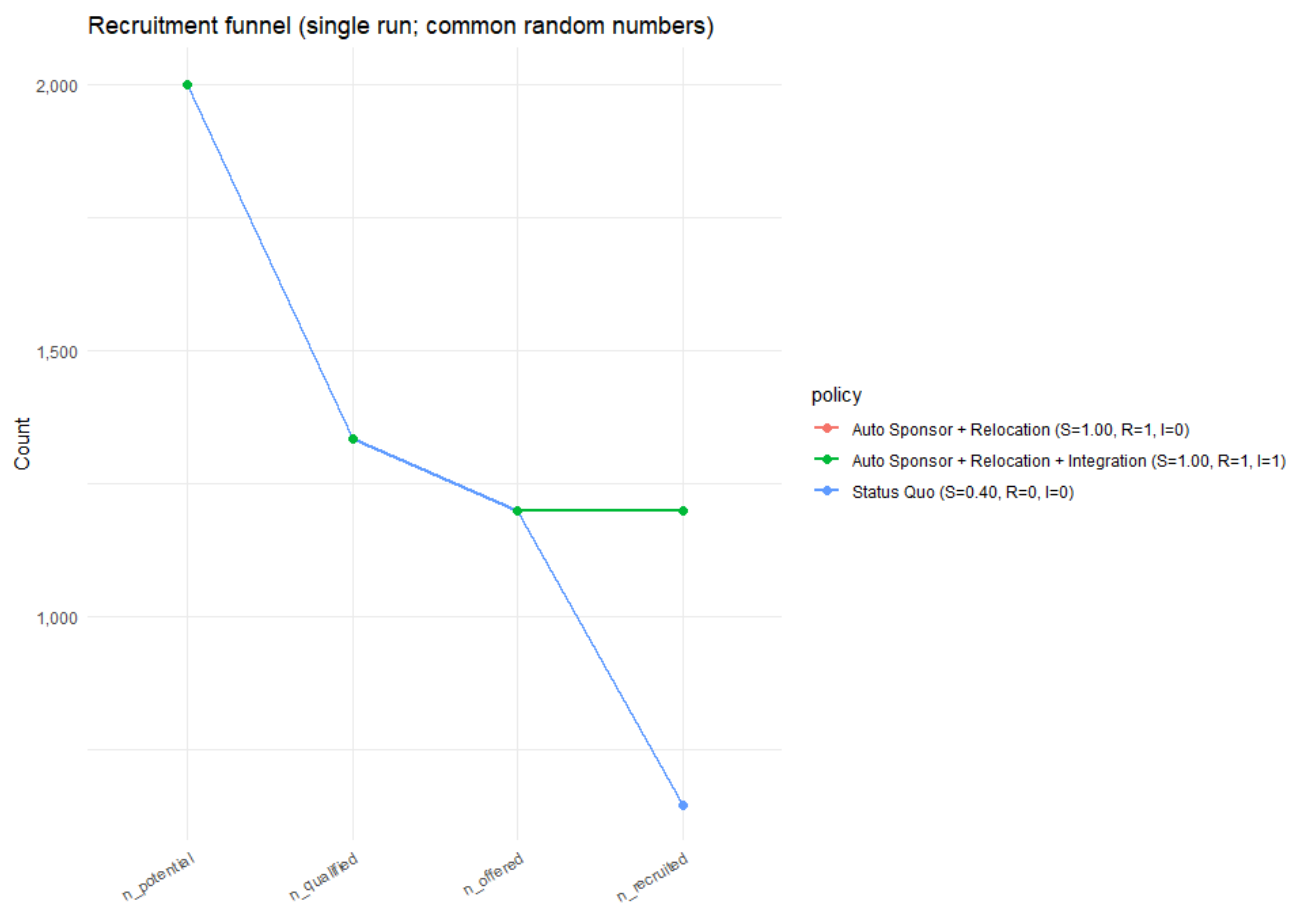
| Policy                                                   | Recruited | Recruited share (of simulated accepted cohort) | Recruited / capacity | Recruited / qualified | Attrition (yr1) | DfE s  |
|----------------------------------------------------------|-----------|------------------------------------------------|----------------------|-----------------------|-----------------|--------|
| S=0.40  <br>None<br>(R=0, I=0)                           | 647       | 0.3235                                         | 0.5392               | 0.4854                | 0.1515          | £3,918 |
| S=0.40  <br>Relocation<br>only (R=1,<br>I=0)             | 647       | 0.3235                                         | 0.5392               | 0.4854                | 0.0757          | £4,268 |
| S=0.40  <br>Integration<br>only (R=0,<br>I=1)            | 647       | 0.3235                                         | 0.5392               | 0.4854                | 0.1128          | £3,918 |
| S=0.40  <br>Relocation<br>+<br>Integration<br>(R=1, I=1) | 647       | 0.3235                                         | 0.5392               | 0.4854                | 0.0572          | £4,268 |

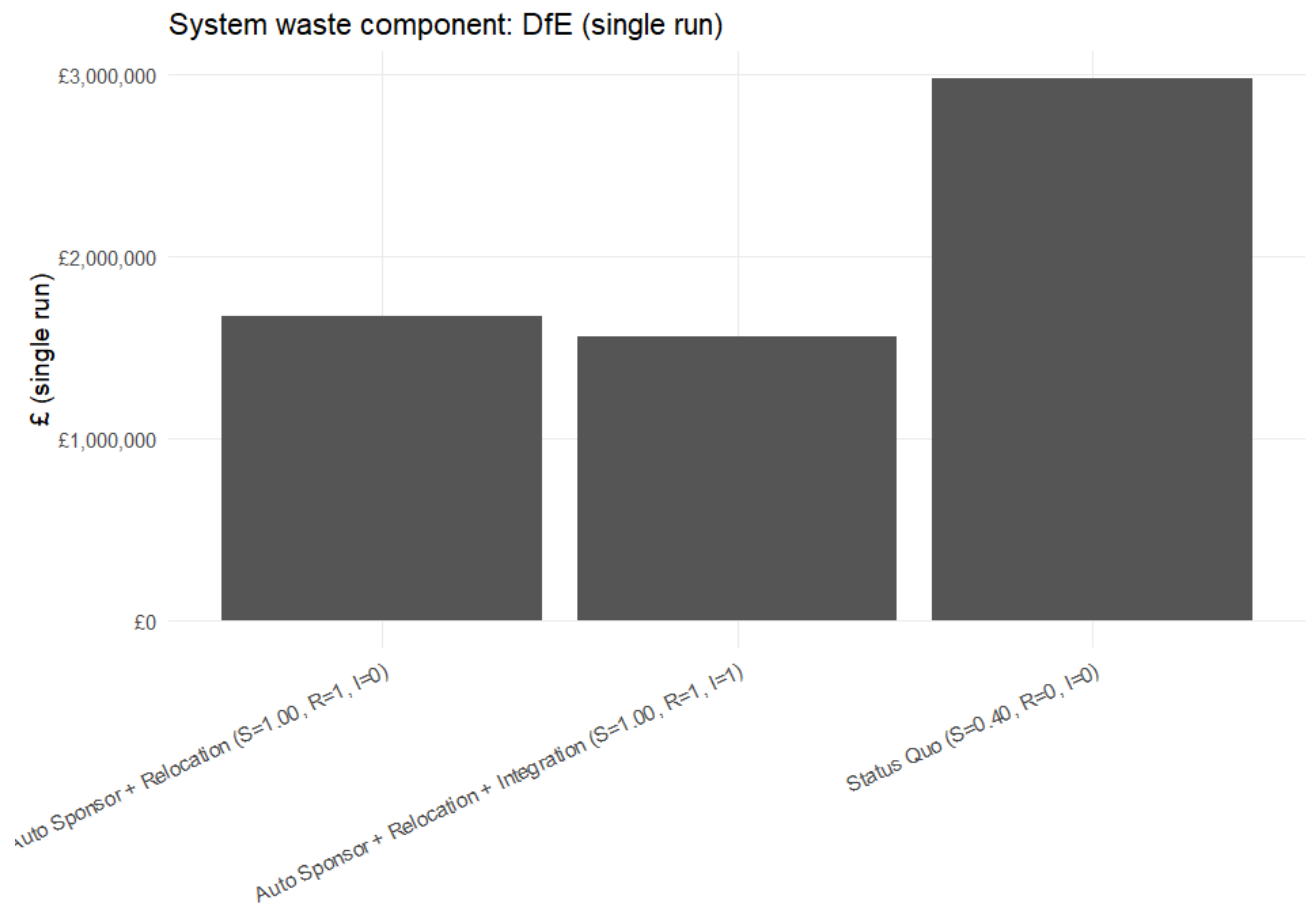
2×2 table (S=0.40) on the same pipeline + common random numbers.

| Policy                         | Recruited | Recruited share (of simulated accepted cohort) | Recruited / capacity | Recruited / qualified | Attrition (yr1) | DfE s  |
|--------------------------------|-----------|------------------------------------------------|----------------------|-----------------------|-----------------|--------|
| S=1.00  <br>None<br>(R=0, I=0) | 1200      | 0.6                                            | 1                    | 0.9002                | 0.1683          | £3,918 |
| S=1.00  <br>Relocation         | 1200      | 0.6                                            | 1                    | 0.9002                | 0.0742          | £4,858 |

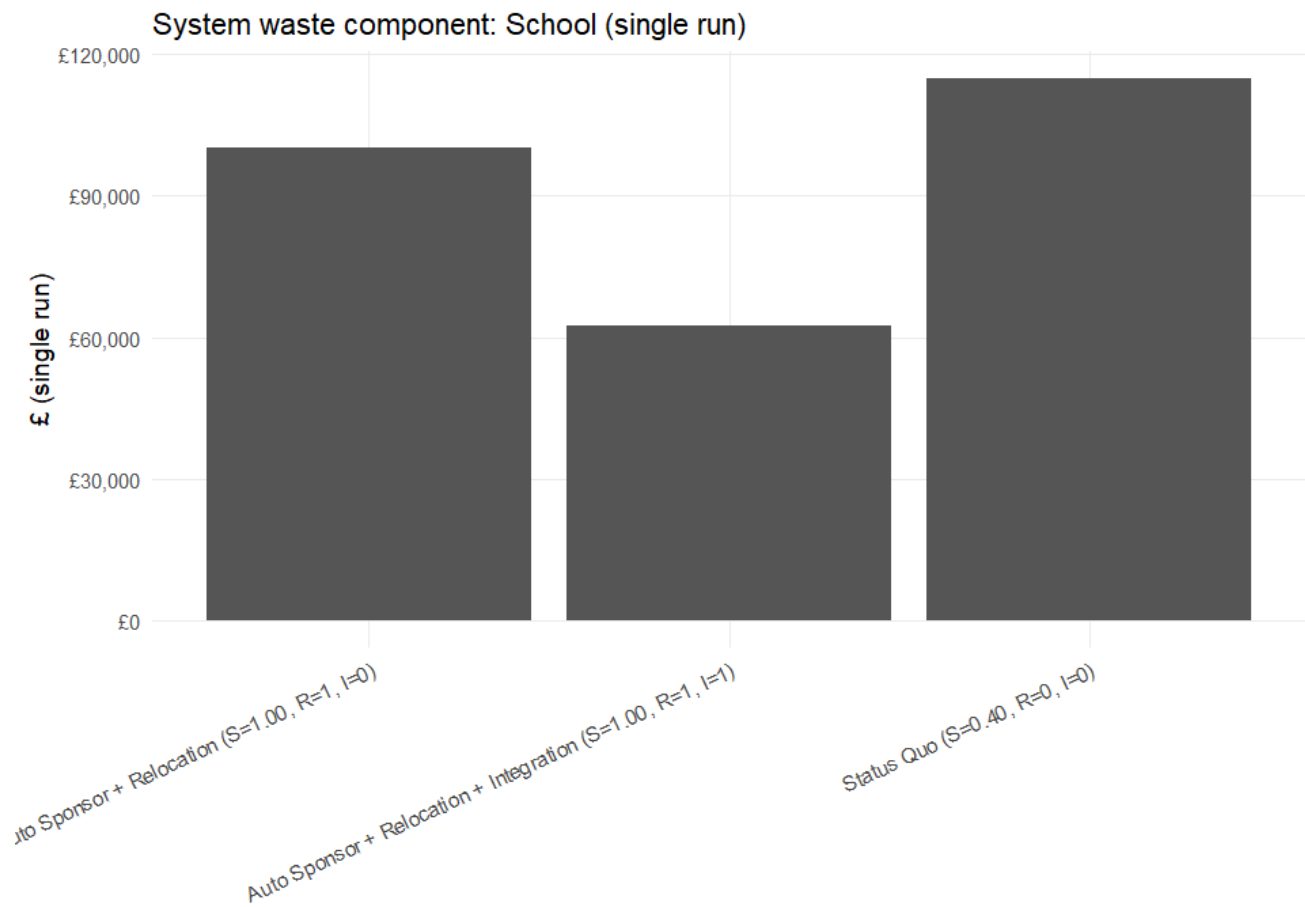
| Policy                                                   | Recruited | Recruited<br>share (of<br>simulated<br>accepted<br>cohort) | Recruited<br>/ capacity | Recruited<br>/<br>qualified | Attrition<br>(yr1) | DfE s  |
|----------------------------------------------------------|-----------|------------------------------------------------------------|-------------------------|-----------------------------|--------------------|--------|
| only (R=1,<br>I=0)                                       |           |                                                            |                         |                             |                    |        |
| S=1.00  <br>Integration<br>only (R=0,<br>I=1)            | 1200      | 0.6                                                        | 1                       | 0.9002                      | 0.1208             | £3,916 |
| S=1.00  <br>Relocation<br>+<br>Integration<br>(R=1, I=1) | 1200      | 0.6                                                        | 1                       | 0.9002                      | 0.0483             | £4,856 |

2x2 table (S=1.00) on the same pipeline + common random numbers.



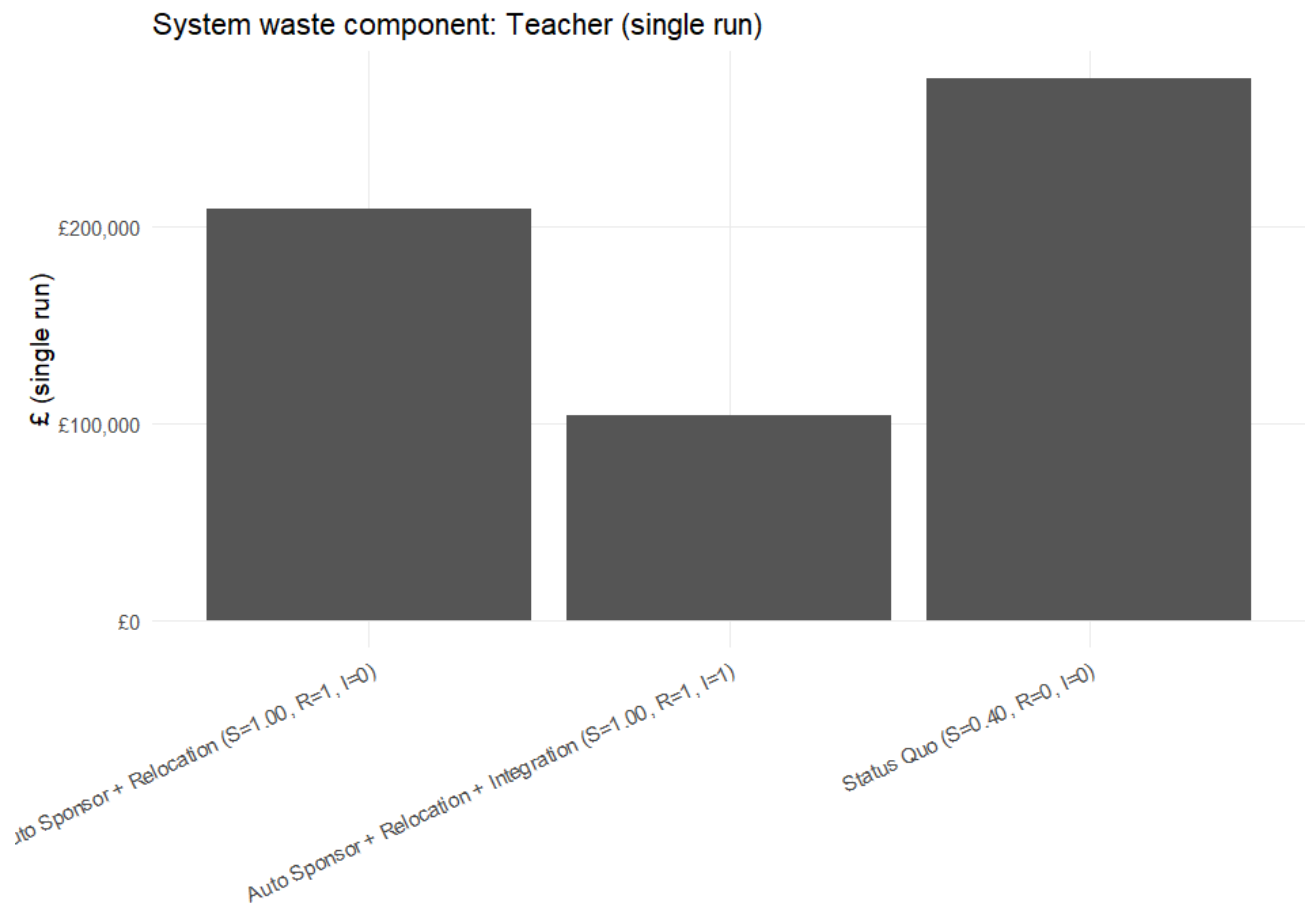


System waste decomposition by policy (single run, CRN): DfE vs School vs Teacher (one plot per component).



System waste decomposition by policy (single run, CRN): DfE vs School vs Teacher (one plot per component).





System waste decomposition by policy (single run, CRN): DfE vs School vs Teacher (one plot per component).

```
# =====
# Block 4 – Monte Carlo replication (R = 5000) with CRN per run
#   Caching: load if schema matches; otherwise run + cache.
#   CI standard: 99% (z_{0.995}).
# =====

stopifnot(
  exists("generate_potential_overseas"),
  exists("add_training_and_offers"),
  exists("assign_training_incentives"),
  exists("apply_policy_and_score"),
  exists("policy_A"),
  exists("policy_B"),
  exists("policy_C")
)

R <- 5000
N0 <- 2000

policy_map <- tibble::tibble(
  policy = c(policy_A$name, policy_B$name, policy_C$name),
```

```

    policy_id = c("A", "B", "C"),
    policy_short = c("Status Quo", "Auto Sponsor + Relocation", "Auto Sponsor + Relocat
)

if (!exists("z_0.995")) z_0.995 <- qnorm(0.995)

mc_summarise <- function(x) {
  x <- x[is.finite(x)]
  n <- length(x)
  if (n == 0) {
    return(tibble::tibble(n = 0L, mean = NA_real_, sd = NA_real_, mcse = NA_real_, ci
  )
  }
  m <- mean(x)
  s <- stats::sd(x)
  se <- s / sqrt(n)
  tibble::tibble(
    n = n,
    mean = m,
    sd = s,
    mcse = se,
    ci_low = m - z_0.995 * se,
    ci_high = m + z_0.995 * se
  )
}

# Formatting helpers (always return character)
fmt_num <- function(x, digits = 3) formatC(x, format = "f", digits = digits)
fmt_ci_num <- function(lo, hi, digits = 3) paste0("[", fmt_num(lo, digits), ", ", fmt

run_one_replication <- function(rep_id, N0 = 2000, params = params) {

  potential <- generate_potential_overseas(
    n = N0,
    probs_overseas_group = probs_overseas_group,
    probs_subject_overseas = probs_subject_overseas,
    probs_degree_overseas_all = probs_degree_overseas_all,
    p_eu_has_euss = params$p_eu_has_euss,
    prob_exceptional = params$prob_exceptional
  )

  # Training completion + offer gate + incentives
  pipeline <- add_training_and_offers(
    df_potential = potential,
    capacity_posts = params$capacity_posts,
    u_qual = runif(nrow(potential)),
    seed_offer = 777 + rep_id
  ) |>
  assign_training_incentives(params = params)

```

```

# Common random numbers across policies within this replication
n_pipe <- nrow(pipeline)
u_sponsor_all <- runif(n_pipe)
u_retention_all <- runif(n_pipe)
u_integration_all <- runif(n_pipe)

res_A <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_A,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)$summary

res_B <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_B,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)$summary

res_C <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_C,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)$summary

dplyr::bind_rows(res_A, res_B, res_C) |>
  dplyr::mutate(
    rep = rep_id,
    total_system_waste = total_dfe_waste + total_school_waste + total_teacher_waste
    retained_year1 = n_recruited * (1 - attrition_rate_year1),
    waste_per_retained = total_system_waste / pmax(retained_year1, 1)
  ) |>
  dplyr::left_join(policy_map, by = "policy") |>
  dplyr::select(rep, policy_id, policy, policy_short, dplyr::everything())
}

# Cache (versioned + schema gated)
mc_path <- cache_path(paste0("mc_abc_R", R))

```

```

required_cols <- c(
  "rep", "policy_id", "policy", "policy_short",
  "n_recruited", "n_qualified", "n_offered",
  "recruitment_rate", "recruited_over_capacity", "recruited_over_qualified",
  "attrition_rate_year1", "share_needs_visa_among_recruited", "share_integration_amon",
  "total_dfe_spend", "total_dfe_waste", "dfe_waste_rate",
  "total_school_spend", "total_school_waste", "school_waste_rate",
  "total_teacher_spend", "total_teacher_waste", "teacher_waste_rate",
  "total_system_waste", "retained_year1", "waste_per_retained"
)

if (csv_schema_matches(mc_path, required_cols)) {
  mc_abc <- readr::read_csv(mc_path, show_col_types = FALSE)
  cat("Loaded cached ", mc_path, "\n", sep = "")
} else {
  set.seed(20250117)

  set.seed(20250117)
mc_abc <- run_simulation_parallel(
  R = R,
  N0 = N0,
  params = params,
  workers = NULL,
  plan_strategy = "multisession",
  progress = FALSE
)

  readr::write_csv(mc_abc, mc_path)
  cat("Ran simulation and cached ", mc_path, "\n", sep = "")
}

```

```
## Loaded cached cache/mc_abc_R5000_v4_99ci_ABF_parallel_20260425.csv
```

```

metrics <- c(
  "n_recruited",
  "recruitment_rate",
  "recruited_over_capacity",
  "recruited_over_qualified",
  "attrition_rate_year1",
  "share_needs_visa_among_recruited",
  "total_dfe_spend",
  "total_dfe_waste",
  "dfe_waste_rate",
  "total_school_spend",
  "total_school_waste",

```

```

"school_waste_rate",
"total_teacher_spend",
"total_teacher_waste",
"teacher_waste_rate",
"total_system_waste",
"waste_per_retained"
)

mc_summary <- mc_abc |>
  dplyr::select(policy_id, policy_short, dplyr::all_of(metrics)) |>
  tidyr::pivot_longer(cols = dplyr::all_of(metrics), names_to = "metric", values_to =
  dplyr::group_by(policy_id, policy_short, metric) |>
  dplyr::group_modify(~ mc_summarise(.x$value)) |>
  dplyr::ungroup()

mc_key_tbl <- mc_summary |>
  dplyr::filter(metric %in% c("n_recruited", "recruitment_rate", "attrition_rate_year1")) |>
  dplyr::mutate(
    mean_fmt = dplyr::case_when(
      metric == "total_system_waste" ~ pound(mean, accuracy = 1),
      metric == "waste_per_retained" ~ pound(mean, accuracy = 0.01),
      metric %in% c("attrition_rate_year1", "recruitment_rate") ~ scales::percent(mean,
      TRUE ~ fmt_num(mean, digits = 0)
    ),
    ci_fmt = dplyr::case_when(
      metric == "total_system_waste" ~ paste0("[", pound(ci_low, accuracy = 1), ", ",
      metric == "waste_per_retained" ~ paste0("[", pound(ci_low, accuracy = 0.01), ", ",
      metric %in% c("attrition_rate_year1", "recruitment_rate") ~ paste0("[", scales::
      TRUE ~ fmt_ci_num(ci_low, ci_high, digits = 0)
    )
  ) |>
  dplyr::select(policy_id, policy_short, metric, n, mean_fmt, ci_fmt, mcse) |>
  dplyr::arrange(policy_id, metric)

knitr::kable(
  mc_key_tbl,
  caption = "Monte Carlo summary (R=5000): mean with 99% CI and MCSE (paired CRN with
)

```

| policy_id | policy_short | metric               | n    | mean_fmt | ci_fmt              |
|-----------|--------------|----------------------|------|----------|---------------------|
| A         | Status Quo   | attrition_rate_year1 | 5000 | 17.15%   | [17.09%,<br>17.20%] |
| A         | Status Quo   | n_recruited          | 5000 | 644      | [643, 644]          |

| policy_id | policy_short                                     | metric               | n    | mean_fmt   | ci_fmt                     |
|-----------|--------------------------------------------------|----------------------|------|------------|----------------------------|
| A         | Status Quo                                       | recruitment_rate     | 5000 | 32.19%     | [32.16%,<br>32.22%]        |
| A         | Status Quo                                       | total_system_waste   | 5000 | £3,487,343 | [£3,477,251<br>£3,497,435] |
| A         | Status Quo                                       | waste_per_retained   | 5000 | £6,548.15  | [£6,526.41,<br>£6,569.89]  |
| B         | Auto<br>Sponsor +<br>Relocation                  | attrition_rate_year1 | 5000 | 8.80%      | [8.77%,<br>8.83%]          |
| B         | Auto<br>Sponsor +<br>Relocation                  | n_recruited          | 5000 | 1200       | [1200, 1200]               |
| B         | Auto<br>Sponsor +<br>Relocation                  | recruitment_rate     | 5000 | 60.00%     | [60.00%,<br>60.00%]        |
| B         | Auto<br>Sponsor +<br>Relocation                  | total_system_waste   | 5000 | £2,167,592 | [£2,159,767<br>£2,175,418] |
| B         | Auto<br>Sponsor +<br>Relocation                  | waste_per_retained   | 5000 | £1,981.36  | [£1,973.99,<br>£1,988.73]  |
| C         | Auto<br>Sponsor +<br>Relocation +<br>Integration | attrition_rate_year1 | 5000 | 6.09%      | [6.06%,<br>6.11%]          |
| C         | Auto<br>Sponsor +<br>Relocation +<br>Integration | n_recruited          | 5000 | 1200       | [1200, 1200]               |
| C         | Auto<br>Sponsor +                                | recruitment_rate     | 5000 | 60.00%     | [60.00%,<br>60.00%]        |

| policy_id | policy_short                                     | metric             | n    | mean_fmt   | ci_fmt                     |
|-----------|--------------------------------------------------|--------------------|------|------------|----------------------------|
|           | Relocation +<br>Integration                      |                    |      |            |                            |
| C         | Auto<br>Sponsor +<br>Relocation +<br>Integration | total_system_waste | 5000 | £1,907,444 | [£1,899,967<br>£1,914,922] |
| C         | Auto<br>Sponsor +<br>Relocation +<br>Integration | waste_per_retained | 5000 | £1,692.96  | [£1,686.20,<br>£1,699.72]  |

Monte Carlo summary (R=5000): mean with 99% CI and MCSE (paired CRN within replication).

```
wide <- mc_abc |>
  dplyr::select(rep, policy_id, waste_per_retained, total_system_waste, attrition_rate) %>
  tidyr::pivot_wider(names_from = policy_id, values_from = c(waste_per_retained, total_system_waste, attrition_rate))

diffs <- wide |>
  dplyr::transmute(
    rep,
    d_waste_B_minus_A = waste_per_retained_B - waste_per_retained_A,
    d_waste_C_minus_A = waste_per_retained_C - waste_per_retained_A,
    d_system_B_minus_A = total_system_waste_B - total_system_waste_A,
    d_system_C_minus_A = total_system_waste_C - total_system_waste_A,
    d_attrition_B_minus_A = attrition_rate_year1_B - attrition_rate_year1_A,
    d_attrition_C_minus_A = attrition_rate_year1_C - attrition_rate_year1_A
  )

diff_summary <- diffs |>
  tidyr::pivot_longer(cols = -rep, names_to = "contrast", values_to = "value") |>
  dplyr::group_by(contrast) |>
  dplyr::group_modify(~ mc_summarise(.x$value)) |>
  dplyr::ungroup()

# IMPORTANT: every branch returns character (no type mixing)
diff_tbl <- diff_summary |>
  dplyr::mutate(
    mean_fmt = dplyr::case_when(
      stringr::str_detect(contrast, "d_system_") ~ pound(mean, accuracy = 1),
      stringr::str_detect(contrast, "d_waste_") ~ pound(mean, accuracy = 0.01),
      stringr::str_detect(contrast, "d_attrition_") ~ scales::percent(mean, accuracy = 0.01)
    )
  )
```

```

    TRUE ~ fmt_num(mean, digits = 3)
  ),
  ci_fmt = dplyr::case_when(
    stringr::str_detect(contrast, "d_system_") ~ paste0("[", pound(ci_low, accuracy
    stringr::str_detect(contrast, "d_waste_") ~ paste0("[", pound(ci_low, accuracy
    stringr::str_detect(contrast, "d_attrition_") ~ paste0("[", scales::percent(ci_
    TRUE ~ fmt_ci_num(ci_low, ci_high, digits = 3)
  )
) |>
dplyr::select(contrast, n, mean_fmt, ci_fmt, mcse) |>
dplyr::arrange(contrast)

knitr::kable(
  diff_tbl,
  caption = "Paired policy effects (B-A and C-A): mean with 99% CI and MCSE (R=5000).
)

```

| contrast              | n    | mean_fmt        | ci_fmt                         | mcse         |
|-----------------------|------|-----------------|--------------------------------|--------------|
| d_attrition_B_minus_A | 5000 | -8.35%          | [-8.39%, -8.30%]               | 0.0001768    |
| d_attrition_C_minus_A | 5000 | -11.06%         | [-11.11%, -11.02%]             | 0.0001810    |
| d_system_B_minus_A    | 5000 | -<br>£1,319,751 | [-£1,326,984, -<br>£1,312,518] | 2808.1053926 |
| d_system_C_minus_A    | 5000 | -<br>£1,579,899 | [-£1,587,330, -<br>£1,572,467] | 2885.1174722 |
| d_waste_B_minus_A     | 5000 | -£4,566.79      | [-£4,584.76, -<br>£4,548.82]   | 6.9770592    |
| d_waste_C_minus_A     | 5000 | -£4,855.19      | [-£4,873.63, -<br>£4,836.75]   | 7.1585179    |

Paired policy effects (B-A and C-A): mean with 99% CI and MCSE (R=5000).

```

## =====
## Internal Validation Pack – invariants + accounting + monotonicity
## Run AFTER Block 4 (requires mc_abc in memory)
## =====

stopifnot(exists("mc_abc"))
suppressPackageStartupMessages({
  library(dplyr)
  library(tidyr)

```



```

library(tibble)
})

strict <- TRUE          # TRUE = stop knit on failure; FALSE = warn + continue
tol <- 1e-10            # numerical tolerance for identities

fail_if <- function(ok, msg, offenders = NULL) {
  if (isTRUE(ok)) return(invisible(TRUE))
  if (!is.null(offenders) && nrow(offenders) > 0) print(offenders)
  if (strict) stop(msg, call. = FALSE) else warning(msg, call. = FALSE)
  invisible(FALSE)
}

# -----
# V1) Basic bounds + finiteness
# -----
need_cols <- c(
  "rep", "policy_id",
  "n_recruited", "n_qualified", "n_offered",
  "recruitment_rate", "recruited_over_capacity", "recruited_over_qualified",
  "attrition_rate_year1",
  "share_needs_visa_among_recruited", "share_integration_among_recruited_needs_visa",
  "total_dfe_spend", "total_dfe_waste", "dfe_waste_rate",
  "total_school_spend", "total_school_waste", "school_waste_rate",
  "total_teacher_spend", "total_teacher_waste", "teacher_waste_rate",
  "total_system_waste", "retained_year1", "waste_per_retained"
)

miss <- setdiff(need_cols, names(mc_abc))
fail_if(length(miss) == 0, paste0("mc_abc missing required cols: ", paste(miss, collapse = ", ")))

bad_bounds <- mc_abc %>%
  filter(
    !is.finite(recruitment_rate) | recruitment_rate < 0 | recruitment_rate > 1 |
    !is.finite(attrition_rate_year1) | attrition_rate_year1 < 0 | attrition_rate_year1 > 1 |
    (!is.na(share_needs_visa_among_recruited) &
      (share_needs_visa_among_recruited < 0 | share_needs_visa_among_recruited > 1)) &
    (!is.na(share_integration_among_recruited_needs_visa) &
      (share_integration_among_recruited_needs_visa < 0 | share_integration_among_recruited_needs_visa > 1)) &
    !is.finite(total_system_waste) | total_system_waste < 0 |
    !is.finite(waste_per_retained) | waste_per_retained < 0
  ) %>%
  select(rep, policy_id, n_recruited, n_qualified, n_offered,
    recruitment_rate, attrition_rate_year1,
    share_needs_visa_among_recruited, share_integration_among_recruited_needs_visa,
    total_system_waste, waste_per_retained) %>%
  head(12)

```

```

fail_if(nrow(bad_bounds) == 0, "V1 failed: bounds/finite/nonnegativity violations.",

# -----
# V2) Accounting identities (must hold row-wise)
# -----

acct <- mc_abc %>%
  mutate(
    chk_system = abs(total_system_waste - (total_dfe_waste + total_school_waste + tot
    chk_retained = abs(retained_year1 - n_recruited * (1 - attrition_rate_year1)),
    chk_wpr = abs(waste_per_retained - total_system_waste / pmax(retained_year1, 1)),
    chk_dfe_rate = abs(dfe_waste_rate - if_else(total_dfe_spend > 0, total_dfe_waste
    chk_school_rate = abs(school_waste_rate - if_else(total_school_spend > 0, total_s
    chk_teacher_rate = abs(teacher_waste_rate - if_else(total_teacher_spend > 0, tota
  )

bad_acct <- acct %>%
  filter(
    chk_system > tol |
    chk_retained > tol |
    chk_wpr > tol |
    (!is.na(chk_dfe_rate) & chk_dfe_rate > tol) |
    (!is.na(chk_school_rate) & chk_school_rate > tol) |
    (!is.na(chk_teacher_rate) & chk_teacher_rate > tol)
  ) %>%
  select(rep, policy_id, chk_system, chk_retained, chk_wpr, chk_dfe_rate, chk_school_
  head(12)

fail_if(nrow(bad_acct) == 0, "V2 failed: accounting identities do not hold.", bad_acc

bad_waste_vs_spend <- mc_abc %>%
  filter(
    total_dfe_waste - total_dfe_spend > tol |
    total_school_waste - total_school_spend > tol |
    total_teacher_waste - total_teacher_spend > tol
  ) %>%
  select(rep, policy_id,
    total_dfe_spend, total_dfe_waste,
    total_school_spend, total_school_waste,
    total_teacher_spend, total_teacher_waste) %>%
  head(12)

fail_if(nrow(bad_waste_vs_spend) == 0, "V2 failed: some waste exceeds spend.", bad_wa

# -----
# V3) Pipeline invariants (within same rep, A/B/C share the same pipeline)
# n_offered and n_qualified should be identical across policies per rep.
# -----

rep_invar <- mc_abc %>%
  group_by(rep) %>%

```

```

summarise(
  span_offered = max(n_offered) - min(n_offered),
  span_qualified = max(n_qualified) - min(n_qualified),
  .groups = "drop"
) %>%
filter(span_offered != 0 | span_qualified != 0) %>%
head(12)

fail_if(
  nrow(rep_invar) == 0,
  "V3 failed: within-rep pipeline invariants violated (n_offered/n_qualified differ a
  rep_invar
)

# -----
# V4) Policy structural invariants under your design
# B and C are both auto-sponsor -> should recruit everyone offered (n_recruited == n_
# C adds integration boost -> attrition_C must be <= attrition_B per rep (CRN).
# B and C share relocation schedule -> DfE spend and Teacher spend should match.
# -----
wide <- mc_abc %>%
  select(
    rep, policy_id,
    n_recruited, n_offered,
    attrition_rate_year1,
    total_dfe_spend, total_teacher_spend,
    share_integration_among_recruited_needs_visa
  ) %>%
  pivot_wider(
    names_from = policy_id,
    values_from = c(
      n_recruited, n_offered,
      attrition_rate_year1,
      total_dfe_spend, total_teacher_spend,
      share_integration_among_recruited_needs_visa
    )
  )

viol <- wide %>%
  filter(
    n_recruited_B != n_offered_B |
    n_recruited_C != n_offered_C |
    n_recruited_C != n_recruited_B |
    n_recruited_B < n_recruited_A |
    n_recruited_C < n_recruited_A |
    attrition_rate_year1_C > attrition_rate_year1_B + 1e-12 |
    abs(total_dfe_spend_B - total_dfe_spend_C) > tol |
    abs(total_teacher_spend_B - total_teacher_spend_C) > tol |
    (!is.na(share_integration_among_recruited_needs_visa_C) &

```

```

    share_integration_among_recruited_needs_visa_C < 1 - 1e-12)
  ) %>%
  head(12)

fail_if(nrow(viol) == 0, "V4 failed: policy invariants/monotonicity violated.", viol)

cat("✓ Internal Validation Pack PASSED (V1-V4).\n")

```

```
## ✓ Internal Validation Pack PASSED (V1-V4).
```

```

## =====
## Edge-case smoke tests (fast)
## Purpose: "does it run + do invariants hold" on tiny/degenerate settings
## =====

stopifnot(exists("run_one_replication"), exists("params"))

edge_cases <- list(
  list(name = "N0=1, capacity=0", N0 = 1, params = modifyList(params, list(capacity_p
  list(name = "N0=1, capacity=1", N0 = 1, params = modifyList(params, list(capacity_p
  list(name = "EU all EUSS (p=1)", N0 = 200, params = modifyList(params, list(p_eu_ha
  list(name = "EU none EUSS (p=0)", N0 = 200, params = modifyList(params, list(p_eu_h
  list(name = "No exceptional quals", N0 = 200, params = modifyList(params, list(prob
  list(name = "All exceptional quals", N0 = 200, params = modifyList(params, list(pro
)

for (ec in edge_cases) {
  out <- run_one_replication(rep_id = 1, N0 = ec$N0, params = ec$params)

  stopifnot(nrow(out) == 3L)
  stopifnot(all(out$n_recruited <= out$n_offered))
  stopifnot(all(out$total_system_waste >= 0))

  # capacity check only if capacity_posts exists in your params
  if (!is.null(ec$params$capacity_posts)) {
    stopifnot(all(out$n_offered <= ec$params$capacity_posts))
  }

  cat("✓ Edge OK:", ec$name, "\n")
}

```

```

## ✓ Edge OK: N0=1, capacity=0
## ✓ Edge OK: N0=1, capacity=1
## ✓ Edge OK: EU all EUSS (p=1)
## ✓ Edge OK: EU none EUSS (p=0)

```

```
## ✓ Edge OK: No exceptional quals
## ✓ Edge OK: All exceptional quals
```

```
# =====
# Block 5 – Report tables from cached MC (no rerun here)
# CI standard: 99% (z_{0.995}).
# =====

if (!exists("z_0.995")) z_0.995 <- qnorm(0.995)

mc_summarise <- function(x) {
  x <- x[is.finite(x)]
  n <- length(x)
  if (n == 0) {
    return(tibble::tibble(n = 0L, mean = NA_real_, sd = NA_real_, mcse = NA_real_, ci_low = NA_real_, ci_high = NA_real_))
  }
  m <- mean(x)
  s <- stats::sd(x)
  se <- s / sqrt(n)
  tibble::tibble(
    n = n,
    mean = m,
    sd = s,
    mcse = se,
    ci_low = m - z_0.995 * se,
    ci_high = m + z_0.995 * se
  )
}

mc_path <- cache_path("mc_abc_R5000")

required_cols <- c(
  "rep", "policy_id", "policy", "policy_short",
  "n_recruited", "n_qualified", "n_offered",
  "recruitment_rate", "recruited_over_capacity", "recruited_over_qualified",
  "attrition_rate_year1", "share_needs_visa_among_recruited", "share_integration_among_recruited",
  "total_dfe_spend", "total_dfe_waste", "dfe_waste_rate",
  "total_school_spend", "total_school_waste", "school_waste_rate",
  "total_teacher_spend", "total_teacher_waste", "teacher_waste_rate",
  "total_system_waste", "retained_year1", "waste_per_retained"
)

if (!csv_schema_matches(mc_path, required_cols)) {
  stop(
    "MC cache missing or schema mismatch: ", mc_path, "\n",
    "Re-knit from Block 4 to regenerate the cache under the current cache_version."
  )
}
```

```
mc_abc <- readr::read_csv(mc_path, show_col_types = FALSE)
cat("✓ Loaded MC cache for reporting: ", mc_path, "\n", sep = "")
```

```
## ✓ Loaded MC cache for reporting: cache/mc_abc_R5000_v4_99ci_ABF_parallel_20260425.
```

```
# -----
# Clean report tables
# -----
```

```
metrics_main <- c(
  "n_recruited",
  "recruitment_rate",
  "recruited_over_capacity",
  "recruited_over_qualified",
  "attrition_rate_year1",
  "share_needs_visa_among_recruited",
  "share_integration_among_recruited_needs_visa",
  "total_dfe_spend",
  "total_dfe_waste",
  "dfe_waste_rate",
  "total_school_spend",
  "total_school_waste",
  "school_waste_rate",
  "total_teacher_spend",
  "total_teacher_waste",
  "teacher_waste_rate",
  "total_system_waste",
  "waste_per_retained"
)
```

```
mc_summary <- mc_abc %>%
  dplyr::select(policy_id, policy_short, dplyr::all_of(metrics_main)) %>%
  tidyr::pivot_longer(cols = dplyr::all_of(metrics_main), names_to = "metric", values_to = "value")
  dplyr::group_by(policy_id, policy_short, metric) %>%
  dplyr::group_modify(~ mc_summarise(.x$value)) %>%
  dplyr::ungroup()
```

```
label_metric <- function(m) {
  dplyr::case_when(
    m == "n_recruited" ~ "Recruited (count)",
    m == "recruitment_rate" ~ "Recruited share (of simulated accepted cohort)",
    m == "recruited_over_capacity" ~ "Recruited / capacity",
    m == "recruited_over_qualified" ~ "Recruited / qualified",
    m == "attrition_rate_year1" ~ "Attrition rate (year 1)",
    m == "share_needs_visa_among_recruited" ~ "Needs-visa share among recruited",
  )
}
```

```

m == "share_integration_among_recruited_needs_visa" ~ "Integration share among vi
m == "total_dfe_spend" ~ "DfE spend (£)",
m == "total_dfe_waste" ~ "DfE waste (£)",
m == "dfe_waste_rate" ~ "DfE waste rate",
m == "total_school_spend" ~ "School spend (£)",
m == "total_school_waste" ~ "School waste (£)",
m == "school_waste_rate" ~ "School waste rate",
m == "total_teacher_spend" ~ "Teacher spend (£)",
m == "total_teacher_waste" ~ "Teacher waste (£)",
m == "teacher_waste_rate" ~ "Teacher waste rate",
m == "total_system_waste" ~ "System waste (£)",
m == "waste_per_retained" ~ "Waste per retained teacher (£)",
TRUE ~ m
)
}

format_mean_ci <- function(metric, mean, lo, hi) {
  if (metric %in% c("total_dfe_spend", "total_dfe_waste", "total_school_spend", "total_s
    "total_teacher_spend", "total_teacher_waste", "total_system_waste"))
    return(paste0(pound(mean, accuracy = 1), " [", pound(lo, accuracy = 1), ", ", pou
  }
  if (metric == "waste_per_retained") {
    return(paste0(pound(mean, accuracy = 0.01), " [", pound(lo, accuracy = 0.01), ",
  }
  if (metric %in% c("recruitment_rate", "attrition_rate_year1", "share_needs_visa_among
    "share_integration_among_recruited_needs_visa", "dfe_waste_rate", "
    return(paste0(scales::percent(mean, accuracy = 0.01), " [",
      scales::percent(lo, accuracy = 0.01), ", ",
      scales::percent(hi, accuracy = 0.01), "]""))
  }
  paste0(round(mean, 3), " [", round(lo, 3), ", ", round(hi, 3), "]"")
}

format_mcse <- function(metric, mcse) {
  if (!is.finite(mcse)) return(NA_character_)
  if (metric %in% c("total_dfe_spend", "total_dfe_waste", "total_school_spend", "total_s
    "total_teacher_spend", "total_teacher_waste", "total_system_waste",
    return(pound(mcse, accuracy = 0.01))
  }
  if (metric %in% c("recruitment_rate", "attrition_rate_year1", "share_needs_visa_among
    "share_integration_among_recruited_needs_visa", "dfe_waste_rate", "
    return(scales::percent(mcse, accuracy = 0.01))
  }
  as.character(round(mcse, 4))
}

tbl_report <- mc_summary %>%
  dplyr::mutate(
    Metric = label_metric(metric),

```

```

`Mean [99% CI]` = purrr::pmap_chr(
  list(metric, mean, ci_low, ci_high),
  ~ format_mean_ci(..1, ..2, ..3, ..4)
),
MCSE = purrr::map2_chr(metric, mcse, format_mcse)
) %>%
dplyr::select(policy_id, policy_short, Metric, `Mean [99% CI]`, MCSE) %>%
dplyr::arrange(policy_id, Metric)

knitr::kable(
  tbl_report,
  caption = "Monte Carlo results (R=5000): mean with 99% CI and MCSE (report table)."
)

```

| policy_id | policy_short | Metric                                        | Mean [99% CI]                             | MCSE      |
|-----------|--------------|-----------------------------------------------|-------------------------------------------|-----------|
| A         | Status Quo   | Attrition rate<br>(year 1)                    | 17.15%<br>[17.09%,<br>17.20%]             | 0.02%     |
| A         | Status Quo   | DfE spend (£)                                 | £4,214,384<br>[£4,202,895,<br>£4,225,874] | £4,460.49 |
| A         | Status Quo   | DfE waste (£)                                 | £3,051,113<br>[£3,041,242,<br>£3,060,984] | £3,832.23 |
| A         | Status Quo   | DfE waste rate                                | 72.40%<br>[72.27%,<br>72.52%]             | 0.05%     |
| A         | Status Quo   | Integration share<br>among visa-<br>recruited | 0.00% [0.00%,<br>0.00%]                   | 0.00%     |
| A         | Status Quo   | Needs-visa<br>share among<br>recruited        | 57.56%<br>[57.49%,<br>57.63%]             | 0.03%     |
| A         | Status Quo   | Recruited (count)                             | 643.755<br>[643.142,<br>644.369]          | 0.2382    |



| policy_id | policy_short | Metric                                         | Mean [99% CI]                       | MCSE      |
|-----------|--------------|------------------------------------------------|-------------------------------------|-----------|
| A         | Status Quo   | Recruited / capacity                           | 0.536 [0.536, 0.537]                | 2e-04     |
| A         | Status Quo   | Recruited / qualified                          | 0.492 [0.492, 0.493]                | 2e-04     |
| A         | Status Quo   | Recruited share (of simulated accepted cohort) | 32.19% [32.16%, 32.22%]             | 0.01%     |
| A         | Status Quo   | School spend (£)                               | £599,185 [£598,238, £600,133]       | £367.70   |
| A         | Status Quo   | School waste (£)                               | £128,462 [£127,959, £128,965]       | £195.41   |
| A         | Status Quo   | School waste rate                              | 21.44% [21.36%, 21.52%]             | 0.03%     |
| A         | Status Quo   | System waste (£)                               | £3,487,343 [£3,477,251, £3,497,435] | £3,917.90 |
| A         | Status Quo   | Teacher spend (£)                              | £1,435,525 [£1,433,256, £1,437,795] | £880.93   |
| A         | Status Quo   | Teacher waste (£)                              | £307,768 [£306,562, £308,974]       | £468.15   |
| A         | Status Quo   | Teacher waste rate                             | 21.44% [21.36%, 21.52%]             | 0.03%     |
| A         | Status Quo   | Waste per retained teacher (£)                 | £6,548.15 [£6,526.41, £6,569.89]    | £8.44     |

| policy_id | policy_short              | Metric                                         | Mean [99% CI]                          | MCSE      |
|-----------|---------------------------|------------------------------------------------|----------------------------------------|-----------|
| B         | Auto Sponsor + Relocation | Attrition rate (year 1)                        | 8.80% [8.77%, 8.83%]                   | 0.01%     |
| B         | Auto Sponsor + Relocation | DfE spend (£)                                  | £5,174,076<br>[£5,159,955, £5,188,197] | £5,482.15 |
| B         | Auto Sponsor + Relocation | DfE waste (£)                                  | £1,787,057<br>[£1,779,437, £1,794,678] | £2,958.56 |
| B         | Auto Sponsor + Relocation | DfE waste rate                                 | 34.57%<br>[34.45%, 34.70%]             | 0.05%     |
| B         | Auto Sponsor + Relocation | Integration share among visa-recruited         | 0.00% [0.00%, 0.00%]                   | 0.00%     |
| B         | Auto Sponsor + Relocation | Needs-visa share among recruited               | 77.23%<br>[77.19%, 77.28%]             | 0.02%     |
| B         | Auto Sponsor + Relocation | Recruited (count)                              | 1200 [1200, 1200]                      | 0         |
| B         | Auto Sponsor + Relocation | Recruited / capacity                           | 1 [1, 1]                               | 0         |
| B         | Auto Sponsor + Relocation | Recruited / qualified                          | 0.918 [0.917, 0.918]                   | 2e-04     |
| B         | Auto Sponsor + Relocation | Recruited share (of simulated accepted cohort) | 60.00%<br>[60.00%, 60.00%]             | 0.00%     |
| B         | Auto Sponsor + Relocation | School spend (£)                               | £1,498,633<br>[£1,497,790, £1,499,476] | £327.25   |

| policy_id | policy_short                            | Metric                         | Mean [99% CI]                          | MCSE      |
|-----------|-----------------------------------------|--------------------------------|----------------------------------------|-----------|
| B         | Auto Sponsor + Relocation               | School waste (£)               | £120,770<br>[£120,276, £121,263]       | £191.55   |
| B         | Auto Sponsor + Relocation               | School waste rate              | 8.06% [8.03%, 8.09%]                   | 0.01%     |
| B         | Auto Sponsor + Relocation               | System waste (£)               | £2,167,592<br>[£2,159,767, £2,175,418] | £3,038.04 |
| B         | Auto Sponsor + Relocation               | Teacher spend (£)              | £3,218,632<br>[£3,216,408, £3,220,857] | £863.72   |
| B         | Auto Sponsor + Relocation               | Teacher waste (£)              | £259,765<br>[£258,636, £260,894]       | £438.34   |
| B         | Auto Sponsor + Relocation               | Teacher waste rate             | 8.07% [8.04%, 8.11%]                   | 0.01%     |
| B         | Auto Sponsor + Relocation               | Waste per retained teacher (£) | £1,981.36<br>[£1,973.99, £1,988.73]    | £2.86     |
| C         | Auto Sponsor + Relocation + Integration | Attrition rate (year 1)        | 6.09% [6.06%, 6.11%]                   | 0.01%     |
| C         | Auto Sponsor + Relocation + Integration | DfE spend (£)                  | £5,174,076<br>[£5,159,955, £5,188,197] | £5,482.15 |
| C         | Auto Sponsor + Relocation + Integration | DfE waste (£)                  | £1,676,138<br>[£1,668,813, £1,683,463] | £2,843.92 |
| C         | Auto Sponsor + Relocation + Integration | DfE waste rate                 | 32.43%<br>[32.31%, 32.55%]             | 0.05%     |

| policy_id | policy_short                                  | Metric                                               | Mean [99% CI]                             | MCSE      |
|-----------|-----------------------------------------------|------------------------------------------------------|-------------------------------------------|-----------|
| C         | Auto Sponsor +<br>Relocation +<br>Integration | Integration share<br>among visa-<br>recruited        | 100.00%<br>[100.00%,<br>100.00%]          | 0.00%     |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | Needs-visa<br>share among<br>recruited               | 77.23%<br>[77.19%,<br>77.28%]             | 0.02%     |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | Recruited (count)                                    | 1200 [1200,<br>1200]                      | 0         |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | Recruited /<br>capacity                              | 1 [1, 1]                                  | 0         |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | Recruited /<br>qualified                             | 0.918 [0.917,<br>0.918]                   | 2e-04     |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | Recruited share<br>(of simulated<br>accepted cohort) | 60.00%<br>[60.00%,<br>60.00%]             | 0.00%     |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | School spend (£)                                     | £1,869,352<br>[£1,868,301,<br>£1,870,404] | £408.20   |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | School waste (£)                                     | £84,897<br>[£84,426,<br>£85,368]          | £182.94   |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | School waste<br>rate                                 | 4.54% [4.52%,<br>4.57%]                   | 0.01%     |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | System waste (£)                                     | £1,907,444<br>[£1,899,967,<br>£1,914,922] | £2,903.04 |

| policy_id | policy_short                                  | Metric                               | Mean [99% CI]                             | MCSE    |
|-----------|-----------------------------------------------|--------------------------------------|-------------------------------------------|---------|
| C         | Auto Sponsor +<br>Relocation +<br>Integration | Teacher spend<br>(£)                 | £3,218,632<br>[£3,216,408,<br>£3,220,857] | £863.72 |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | Teacher waste<br>(£)                 | £146,409<br>[£145,549,<br>£147,269]       | £333.94 |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | Teacher waste<br>rate                | 4.55% [4.52%,<br>4.58%]                   | 0.01%   |
| C         | Auto Sponsor +<br>Relocation +<br>Integration | Waste per<br>retained teacher<br>(£) | £1,692.96<br>[£1,686.20,<br>£1,699.72]    | £2.62   |

Monte Carlo results (R=5000): mean with 99% CI and MCSE (report table).

```
# =====
# Appendix A – Robustness: sweep p_eu_has_euss with R=5000
# -----
# For each p in grid:
#   - run R replications with CRN across policies within replication
#   - store MC means (and MCSEs) for key outcomes
#   - write results to CSV for reproducible plotting (versioned + schema-gated)
# =====

stopifnot(
  exists("generate_potential_overseas"),
  exists("add_training_and_offers"),
  exists("assign_training_incentives"),
  exists("apply_policy_and_score"),
  exists("policy_A"),
  exists("policy_B"),
  exists("policy_C"),
  exists("probs_overseas_group"),
  exists("probs_subject_overseas"),
  exists("probs_degree_overseas_all"),
  exists("params"),
  exists("cache_path"),
  exists("csv_schema_matches")
)
```

```

need_params <- c("prob_exceptional", "capacity_posts", "relocation_subjects")
missing_params <- setdiff(need_params, names(params))
if (length(missing_params) > 0) stop("Missing params: ", paste(missing_params, collapse = ", "))

R <- 5000
N0 <- 2000
p_grid <- seq(0.2, 0.9, by = 0.1)

robust_file <- cache_path(paste0("robust_p_eu_has_euss_R", R))
robust_required_cols <- c("p_eu_has_euss", "policy_id", "policy_short", "metric", "n", "mcse")

policy_map <- tibble::tibble(
  policy = c(policy_A$name, policy_B$name, policy_C$name),
  policy_id = c("A", "B", "C"),
  policy_short = c("Status quo", "Auto sponsor + relocation", "Auto sponsor + relocation")
)

mc_summarise <- function(x) {
  x <- x[is.finite(x)]
  n <- length(x)
  if (n == 0) {
    return(tibble::tibble(n = 0L, mean = NA_real_, sd = NA_real_, mcse = NA_real_))
  }
  m <- mean(x)
  s <- stats::sd(x)
  tibble::tibble(n = n, mean = m, sd = s, mcse = s / sqrt(n))
}

# One replication at a given p_eu_has_euss (CRN shared across policies inside rep)
run_one_rep <- function(rep_id, p_eu, N0 = 2000, params = params) {

  potential <- generate_potential_overseas(
    n = N0,
    probs_overseas_group = probs_overseas_group,
    probs_subject_overseas = probs_subject_overseas,
    probs_degree_overseas_all = probs_degree_overseas_all,
    p_eu_has_euss = p_eu,
    prob_exceptional = params$prob_exceptional
  )

  pipeline <- add_training_and_offers(
    df_potential = potential,
    capacity_posts = params$capacity_posts,
    u_qual = runif(nrow(potential)),
    seed_offer = 777 + rep_id
  ) |>
  assign_training_incentives(params = params)

  n_pipe <- nrow(pipeline)

```

```

u_sponsor_all <- runif(n_pipe)
u_retention_all <- runif(n_pipe)
u_integration_all <- runif(n_pipe)

res_A <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_A,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)$summary

res_B <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_B,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)$summary

res_C <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_C,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)$summary

dplyr::bind_rows(res_A, res_B, res_C) |>
  dplyr::mutate(
    rep = rep_id,
    p_eu_has_euss = p_eu,
    total_system_waste = total_dfe_waste + total_school_waste + total_teacher_waste
    retained_year1 = n_recruited * (1 - attrition_rate_year1),
    waste_per_retained = total_system_waste / pmax(retained_year1, 1)
  ) |>
  dplyr::left_join(policy_map, by = "policy") |>
  dplyr::select(p_eu_has_euss, rep, policy_id, policy_short, dplyr::everything())
}

# Build (or load) robustness table (schema-gated)
if (csv_schema_matches(robust_file, robust_required_cols)) {

```

```

robust <- readr::read_csv(robust_file, show_col_types = FALSE)
cat("Loaded cached ", robust_file, "\n", sep = "")

} else {

  set.seed(20250119)

  out_list <- vector("list", length(p_grid))

  for (i in seq_along(p_grid)) {

    p_val <- p_grid[i]

    res_list <- vector("list", R)
    for (r in seq_len(R)) {
      res_list[[r]] <- run_one_rep(r, p_val, N0 = N0, params = params)
    }

    mc <- dplyr::bind_rows(res_list)

    # Metrics: explicit and stable (avoid naming collisions)
    metrics <- c(
      "recruitment_rate",
      "recruited_over_capacity",
      "recruited_over_qualified",
      "attrition_rate_year1",
      "share_needs_visa_among_recruited",
      "total_dfe_waste",
      "total_teacher_waste",
      "total_system_waste",
      "waste_per_retained"
    )
    metrics <- intersect(metrics, names(mc))

    summ <- mc |>
      dplyr::select(p_eu_has_euss, policy_id, policy_short, dplyr::all_of(metrics)) |
      tidyr::pivot_longer(cols = dplyr::all_of(metrics), names_to = "metric", values_
      dplyr::group_by(p_eu_has_euss, policy_id, policy_short, metric) |>
      dplyr::group_modify(~ mc_summarise(.x$value)) |>
      dplyr::ungroup()

    out_list[[i]] <- summ
  }

  robust <- dplyr::bind_rows(out_list)
  readr::write_csv(robust, robust_file)
  cat("Ran robustness sweep and cached ", robust_file, "\n", sep = "")
}

```



```
## Loaded cached cache/robust_p_eu_has_euss_R5000_v4_99ci_ABF_parallel_20260425.csv
```

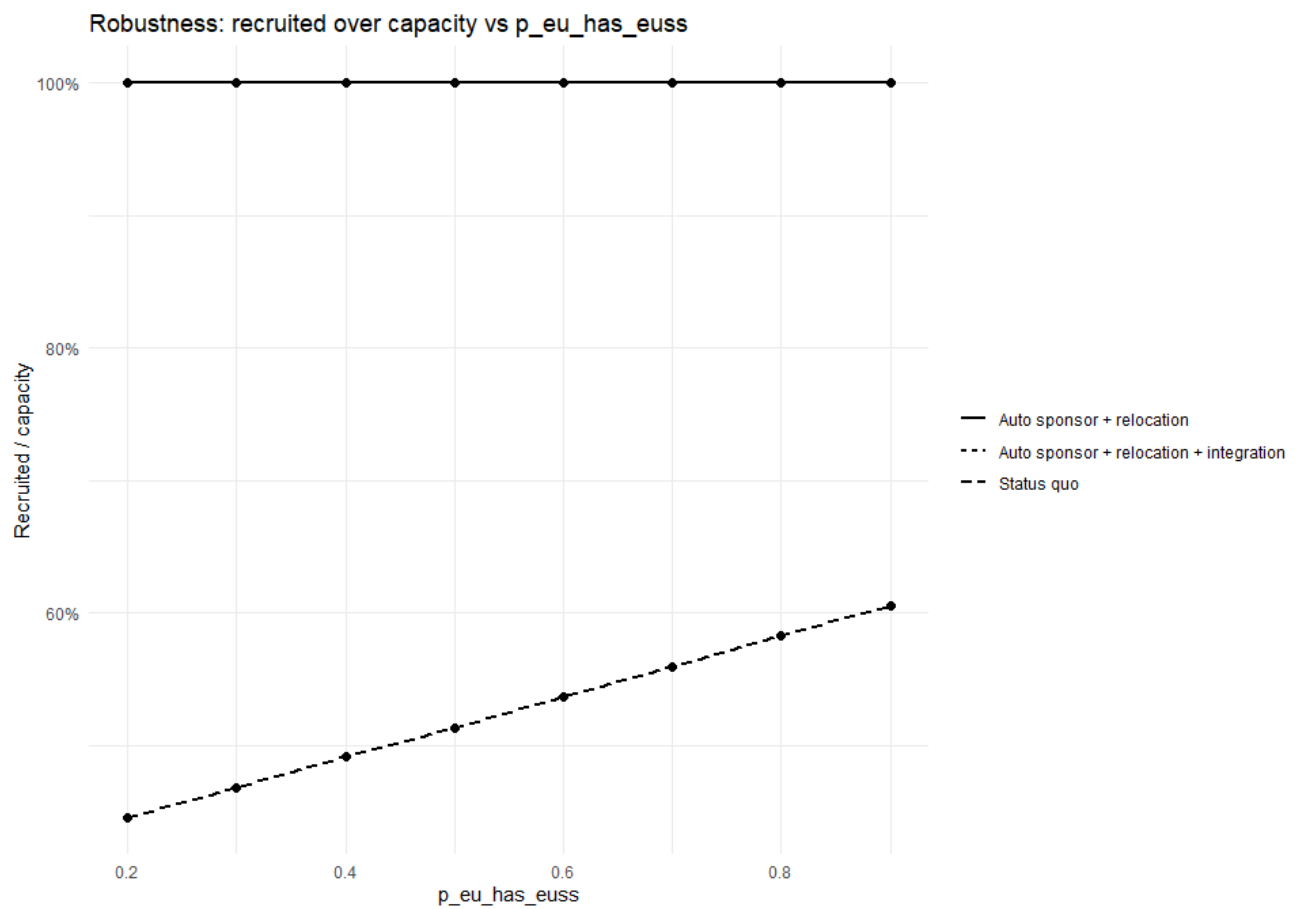
```
# Convenience subsets for plotting
robust_means <- robust |>
  dplyr::select(p_eu_has_euss, policy_short, metric, mean)

plot_line <- function(df, ylab, title, y_formatter = NULL) {
  p <- ggplot(df, aes(x = p_eu_has_euss, y = mean, linetype = policy_short, group = p_eu_has_euss)) +
    geom_line(linewidth = 1, colour = "black") +
    geom_point(size = 2, colour = "black") +
    labs(x = "p_eu_has_euss", y = ylab, title = title) +
    theme_minimal() +
    theme(legend.title = element_blank())

  if (!is.null(y_formatter)) {
    p <- p + scale_y_continuous(labels = y_formatter)
  }
  p
}
```

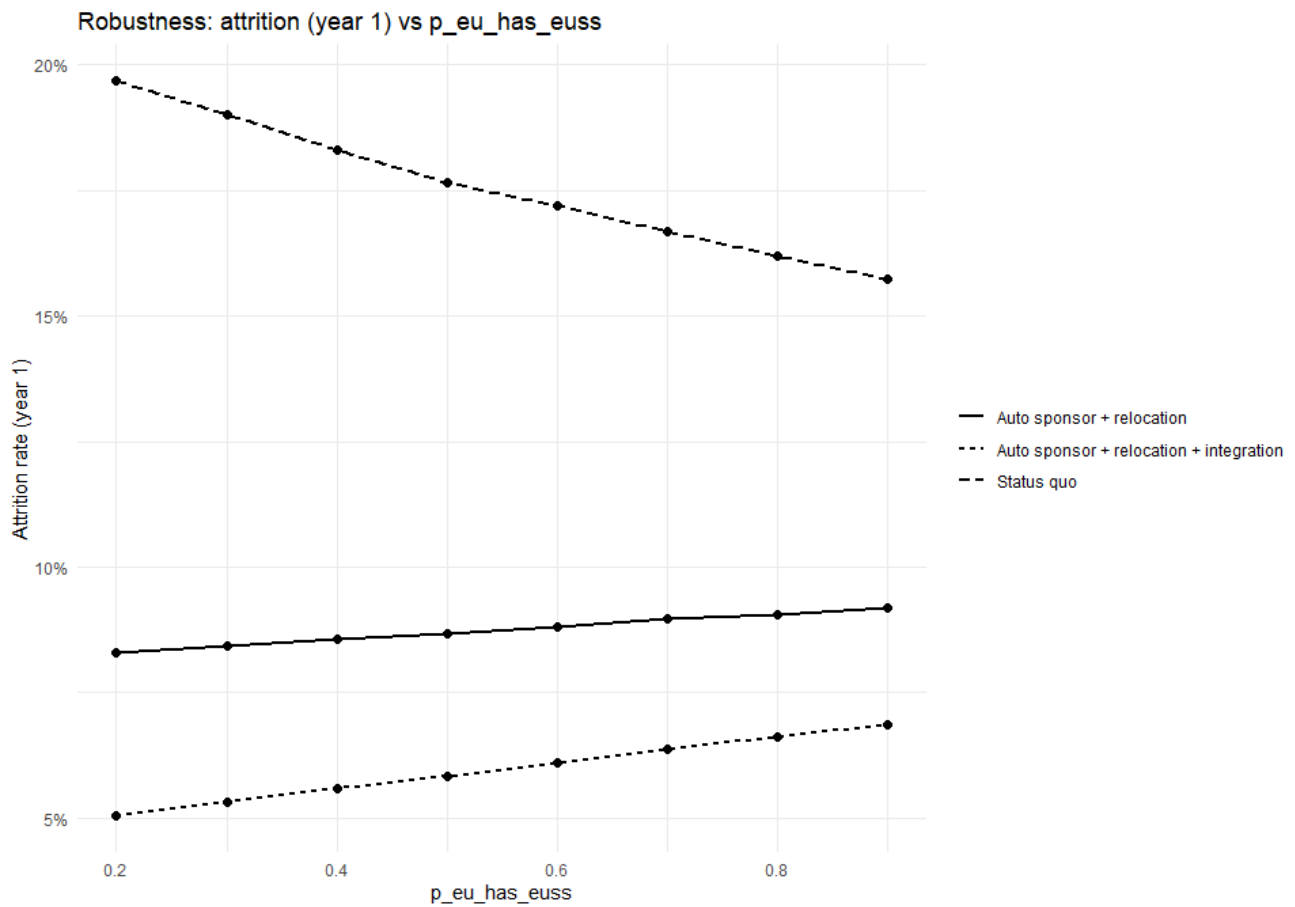
```
df <- robust_means |>
  dplyr::filter(metric == "recruited_over_capacity")

plot_line(
  df,
  ylab = "Recruited / capacity",
  title = "Robustness: recruited over capacity vs p_eu_has_euss",
  y_formatter = scales::percent_format(accuracy = 1)
)
```



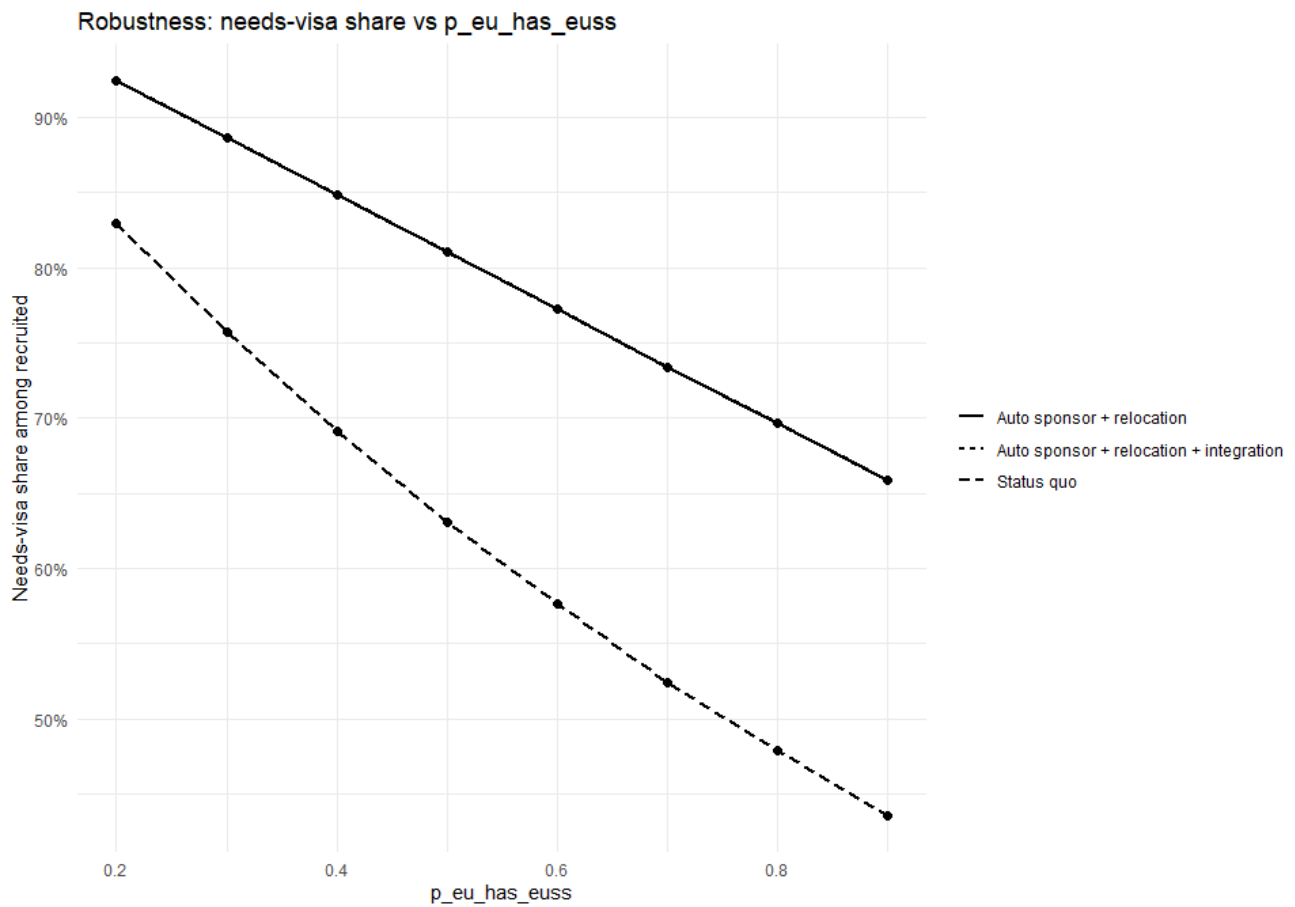
```
df <- robust_means |>
  dplyr::filter(metric == "attrition_rate_year1")

plot_line(
  df,
  ylab = "Attrition rate (year 1)",
  title = "Robustness: attrition (year 1) vs p_eu_has_euss",
  y_formatter = scales::percent_format(accuracy = 1)
)
```



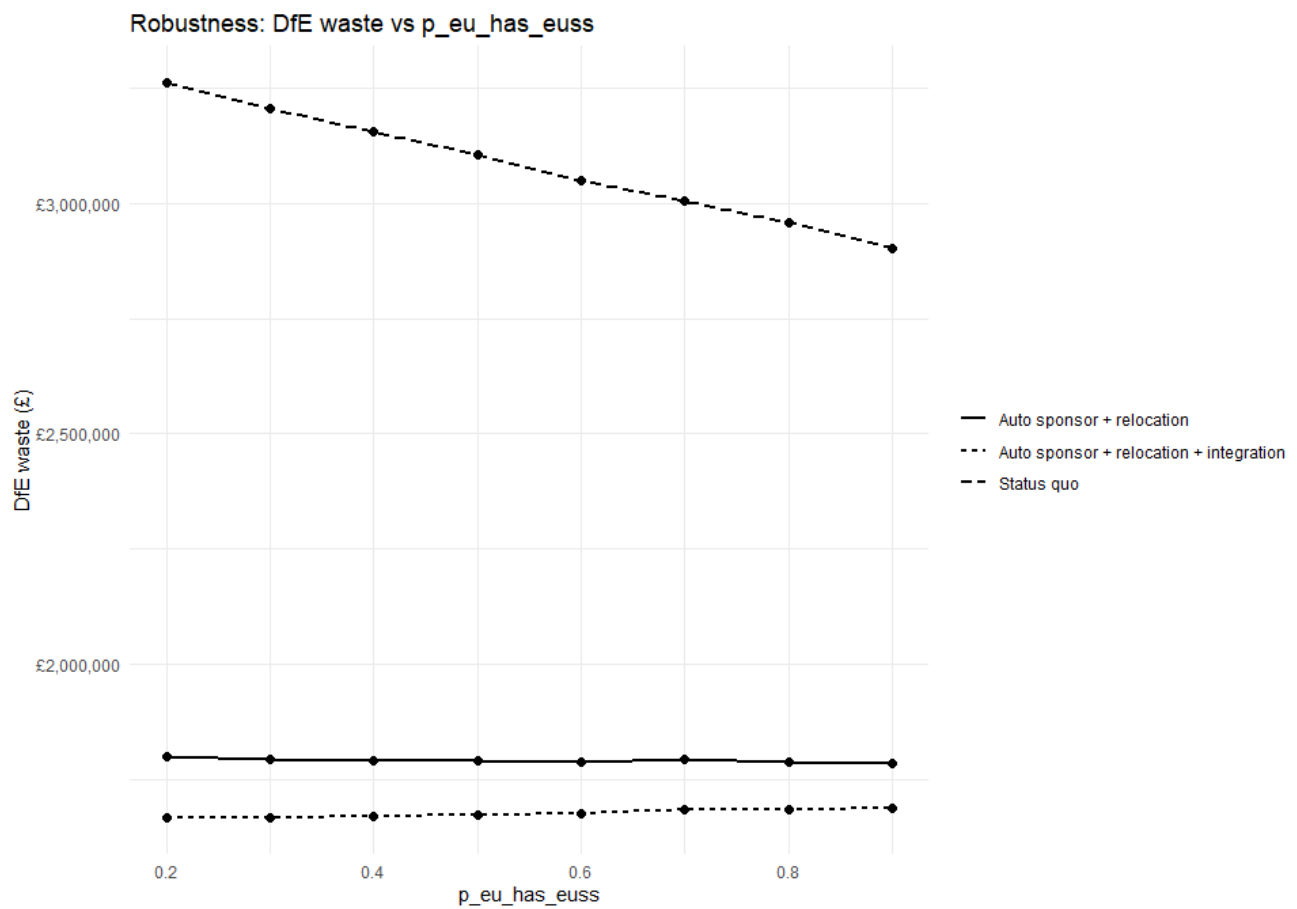
```
df <- robust_means |>
  dplyr::filter(metric == "share_needs_visa_among_recruited")

plot_line(
  df,
  ylab = "Needs-visa share among recruited",
  title = "Robustness: needs-visa share vs p_eu_has_euss",
  y_formatter = scales::percent_format(accuracy = 1)
)
```



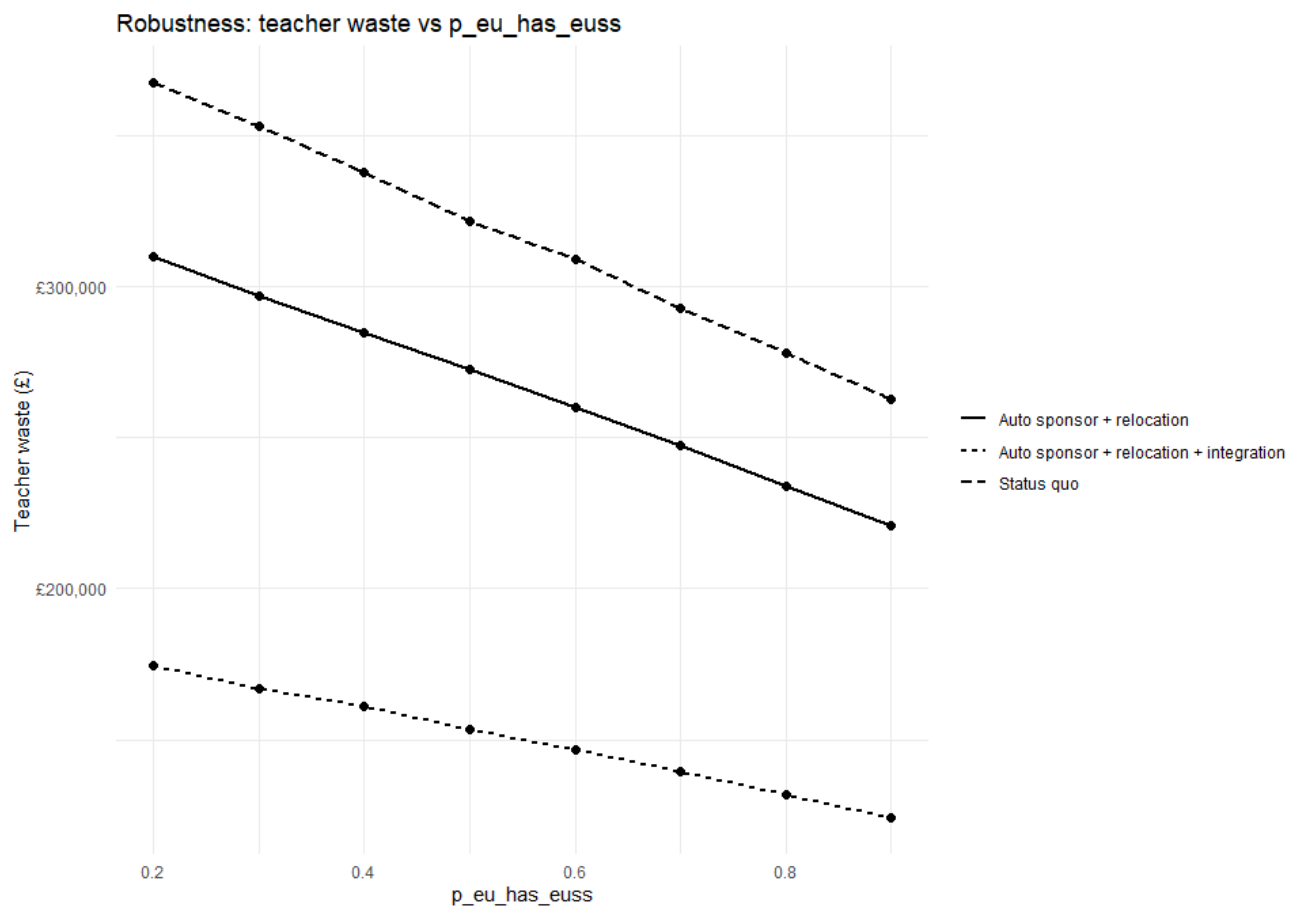
```
df <- robust_means |>
  dplyr::filter(metric == "total_dfe_waste")

plot_line(
  df,
  ylab = "DfE waste (£)",
  title = "Robustness: DfE waste vs p_eu_has_euss",
  y_formatter = pound
)
```



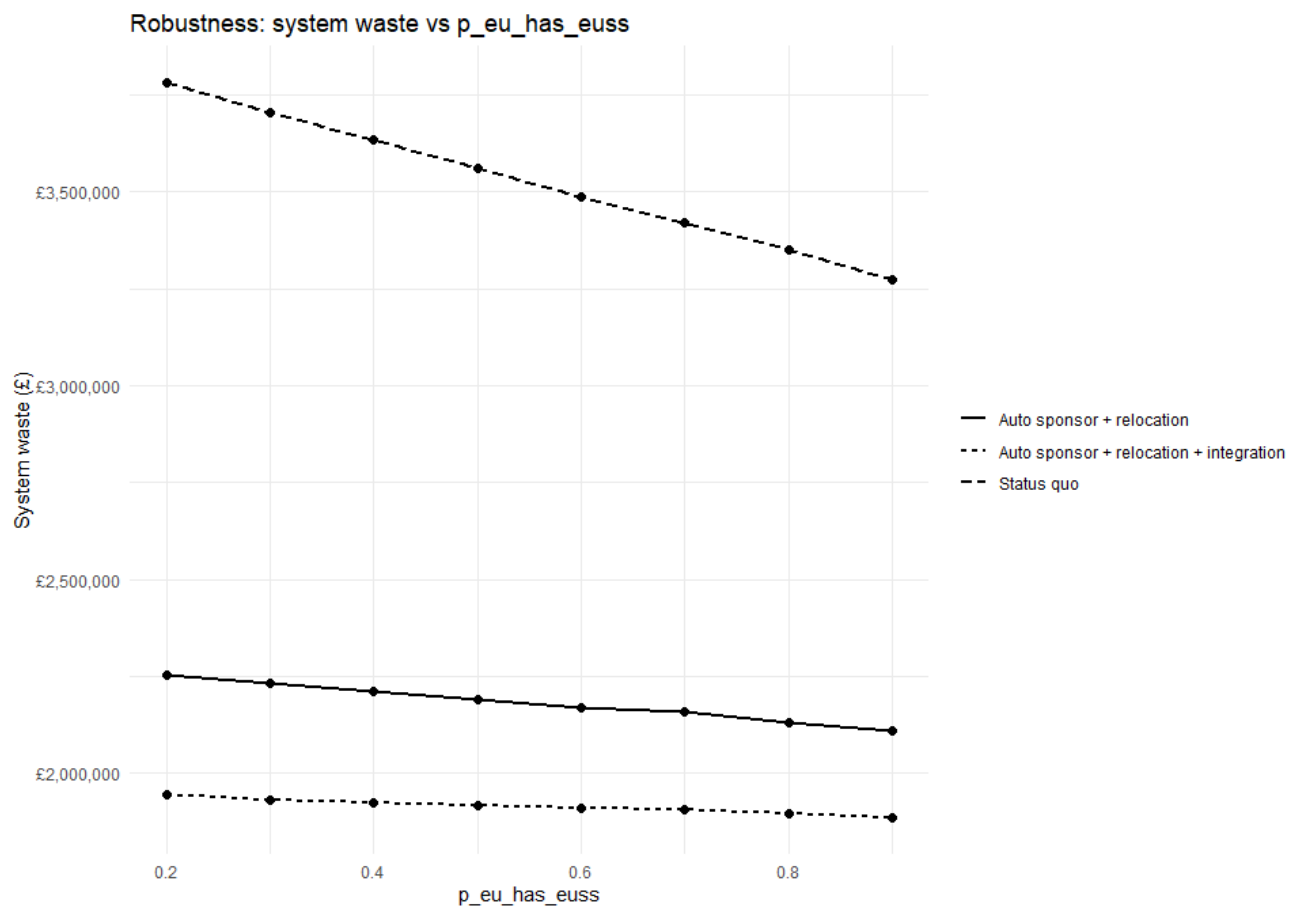
```
df <- robust_means |>
  dplyr::filter(metric == "total_teacher_waste")

plot_line(
  df,
  ylab = "Teacher waste (£)",
  title = "Robustness: teacher waste vs p_eu_has_euss",
  y_formatter = pound
)
```



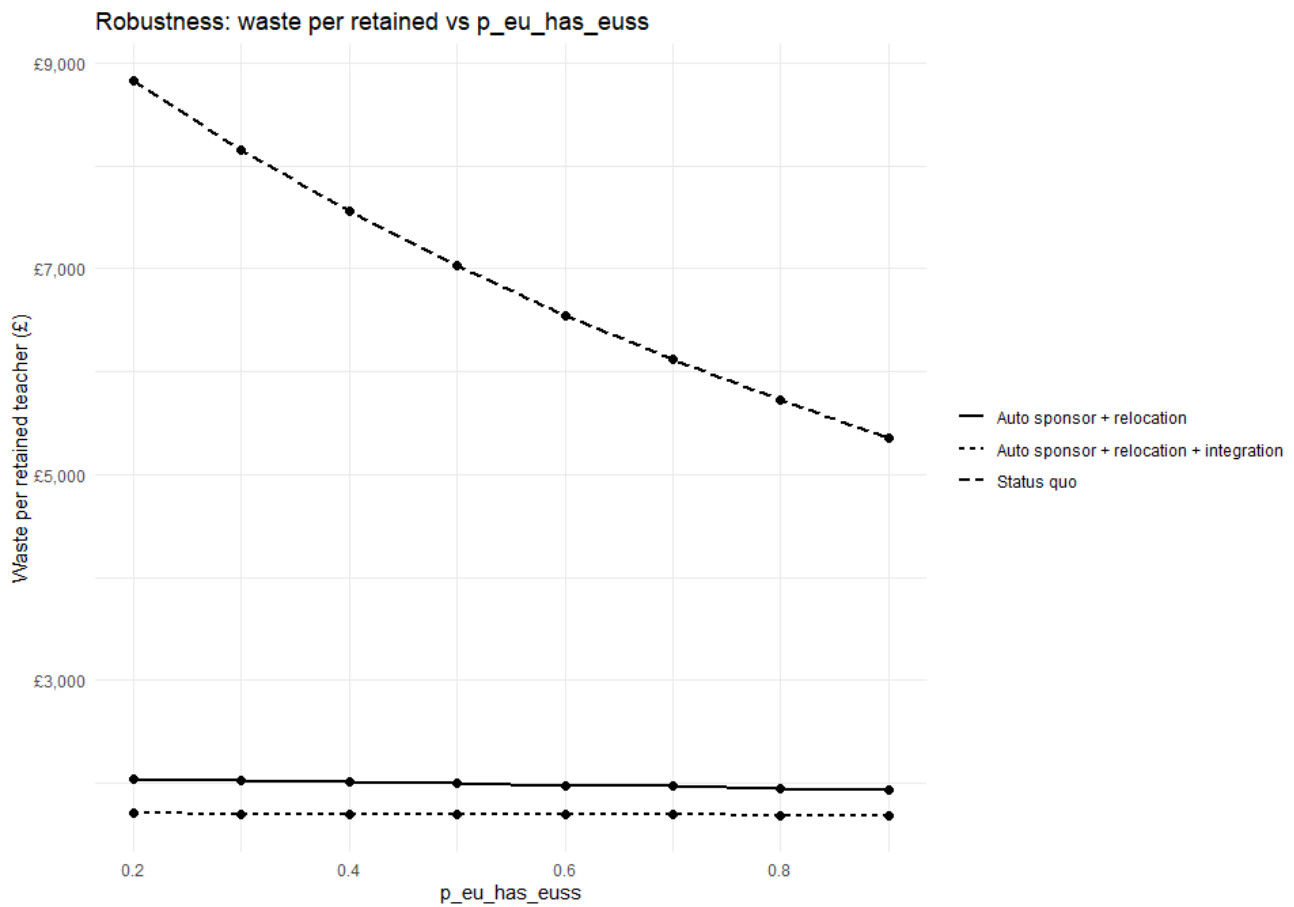
```
df <- robust_means |>
  dplyr::filter(metric == "total_system_waste")

plot_line(
  df,
  ylab = "System waste (£)",
  title = "Robustness: system waste vs p_eu_has_euss",
  y_formatter = pound
)
```



```
df <- robust_means |>
  dplyr::filter(metric == "waste_per_retained")

plot_line(
  df,
  ylab = "Waste per retained teacher (£)",
  title = "Robustness: waste per retained vs p_eu_has_euss",
  y_formatter = pound
)
```



```
# =====
# Appendix B – Robustness: sweep p_eu_has_euss (R=5000)
# Focus: DfE waste among non-visa candidates only: dfe_waste[!needs_visa]
# =====

stopifnot(
  exists("generate_potential_overseas"),
  exists("add_training_and_offers"),
  exists("assign_training_incentives"),
  exists("apply_policy_and_score"),
  exists("policy_A"),
  exists("policy_B"),
  exists("policy_C"),
  exists("probs_overseas_group"),
  exists("probs_subject_overseas"),
  exists("probs_degree_overseas_all"),
  exists("params"),
  exists("cache_path"),
  exists("csv_schema_matches")
)

R <- 5000
N0 <- 2000
p_grid <- seq(0.2, 0.9, by = 0.1)
```



```

robustB_file <- cache_path(paste0("robust_nonvisa_dfe_waste_R", R))
robustB_required_cols <- c("p_eu_has_euss", "policy_id", "policy_short", "metric", "n", "m")

policy_map <- tibble::tibble(
  policy = c(policy_A$name, policy_B$name, policy_C$name),
  policy_id = c("A", "B", "C"),
  policy_short = c("Status quo", "Auto sponsor + relocation", "Auto sponsor + relocation")
)

mc_summarise <- function(x) {
  x <- x[is.finite(x)]
  n <- length(x)
  if (n == 0) return(tibble::tibble(n = 0L, mean = NA_real_, sd = NA_real_, mcse = NA_real_))
  m <- mean(x)
  s <- stats::sd(x)
  tibble::tibble(n = n, mean = m, sd = s, mcse = s / sqrt(n))
}

run_one_rep_B <- function(rep_id, p_eu, N0 = 2000, params = params) {

  potential <- generate_potential_overseas(
    n = N0,
    probs_overseas_group = probs_overseas_group,
    probs_subject_overseas = probs_subject_overseas,
    probs_degree_overseas_all = probs_degree_overseas_all,
    p_eu_has_euss = p_eu,
    prob_exceptional = params$prob_exceptional
  )

  pipeline <- add_training_and_offers(
    df_potential = potential,
    capacity_posts = params$capacity_posts,
    u_qual = runif(nrow(potential)),
    seed_offer = 777 + rep_id
  ) |>
    assign_training_incentives(params = params)

  n_pipe <- nrow(pipeline)
  u_sponsor_all <- runif(n_pipe)
  u_retention_all <- runif(n_pipe)
  u_integration_all <- runif(n_pipe)

  res_A <- apply_policy_and_score(
    df_pipeline = pipeline,
    policy = policy_A,
    relocation_subjects = params$relocation_subjects,
    u_integration = u_integration_all,
    u_sponsor = u_sponsor_all,
    u_retention = u_retention_all,

```

```

    params = params
  )$summary

res_B <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_B,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)$summary

res_C <- apply_policy_and_score(
  df_pipeline = pipeline,
  policy = policy_C,
  relocation_subjects = params$relocation_subjects,
  u_integration = u_integration_all,
  u_sponsor = u_sponsor_all,
  u_retention = u_retention_all,
  params = params
)$summary

dplyr::bind_rows(res_A, res_B, res_C) |>
  dplyr::mutate(
    rep = rep_id,
    p_eu_has_euss = p_eu,
    total_system_waste = total_dfe_waste + total_school_waste + total_teacher_waste
    retained_year1 = n_recruited * (1 - attrition_rate_year1),
    waste_per_retained = total_system_waste / pmax(retained_year1, 1)
  ) |>
  dplyr::left_join(policy_map, by = "policy") |>
  dplyr::select(p_eu_has_euss, rep, policy_id, policy_short, dplyr::everything())
}

if (csv_schema_matches(robustB_file, robustB_required_cols)) {

  robustB <- readr::read_csv(robustB_file, show_col_types = FALSE)
  cat("Loaded cached ", robustB_file, "\n", sep = "")

} else {

  set.seed(20250119)

  out_list <- vector("list", length(p_grid))

  for (i in seq_along(p_grid)) {

    p_val <- p_grid[i]

```

```

res_list <- vector("list", R)
for (r in seq_len(R)) {
  res_list[[r]] <- run_one_rep_B(r, p_val, N0 = N0, params = params)
}

mc <- dplyr::bind_rows(res_list)

metrics <- c(
  "recruitment_rate",
  "recruited_over_capacity",
  "attrition_rate_year1",
  "share_needs_visa_among_recruited",
  "total_dfe_waste_nonvisa",
  "total_system_waste",
  "waste_per_retained"
)
metrics <- intersect(metrics, names(mc))

summ <- mc |>
  dplyr::select(p_eu_has_euss, policy_id, policy_short, dplyr::all_of(metrics)) |
  tidyr::pivot_longer(cols = dplyr::all_of(metrics), names_to = "metric", values_
  dplyr::group_by(p_eu_has_euss, policy_id, policy_short, metric) |>
  dplyr::group_modify(~ mc_summarise(.x$value)) |>
  dplyr::ungroup()

out_list[[i]] <- summ
}

robustB <- dplyr::bind_rows(out_list)
readr::write_csv(robustB, robustB_file)
cat("Ran robustness sweep and cached ", robustB_file, "\n", sep = "")
}

```

```
## Loaded cached cache/robust_nonvisa_dfe_waste_R5000_v4_99ci_ABF_parallel_20260425.c
```

```

robustB_means <- robustB |>
  dplyr::select(p_eu_has_euss, policy_short, metric, mean)

```

```

df <- robustB_means |>
  dplyr::filter(metric == "total_dfe_waste_nonvisa")

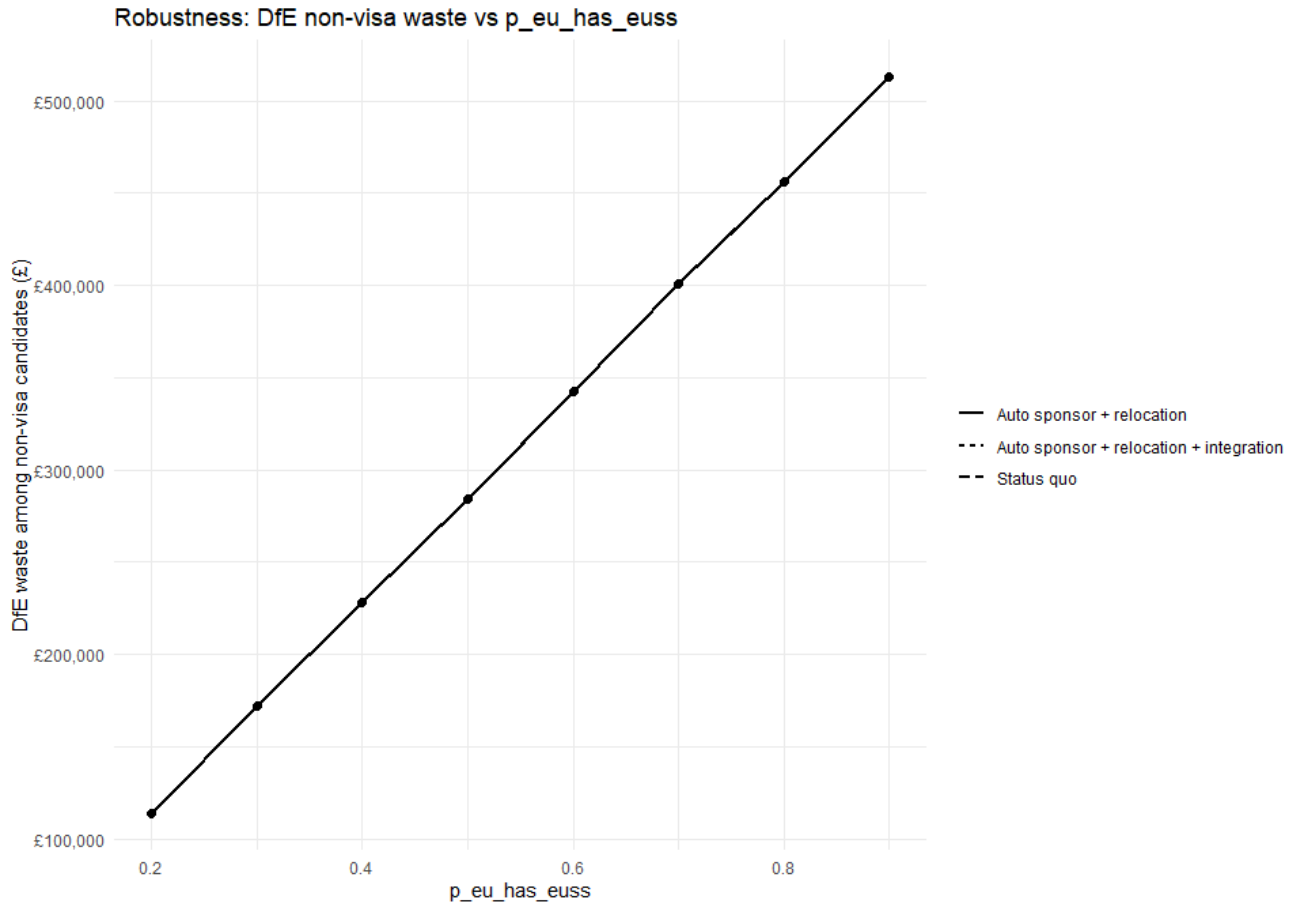
plot_line(
  df,
  ylab = "DfE waste among non-visa candidates (£)",

```

```

title = "Robustness: DfE non-visa waste vs p_eu_has_euss",
y_formatter = pound
)

```

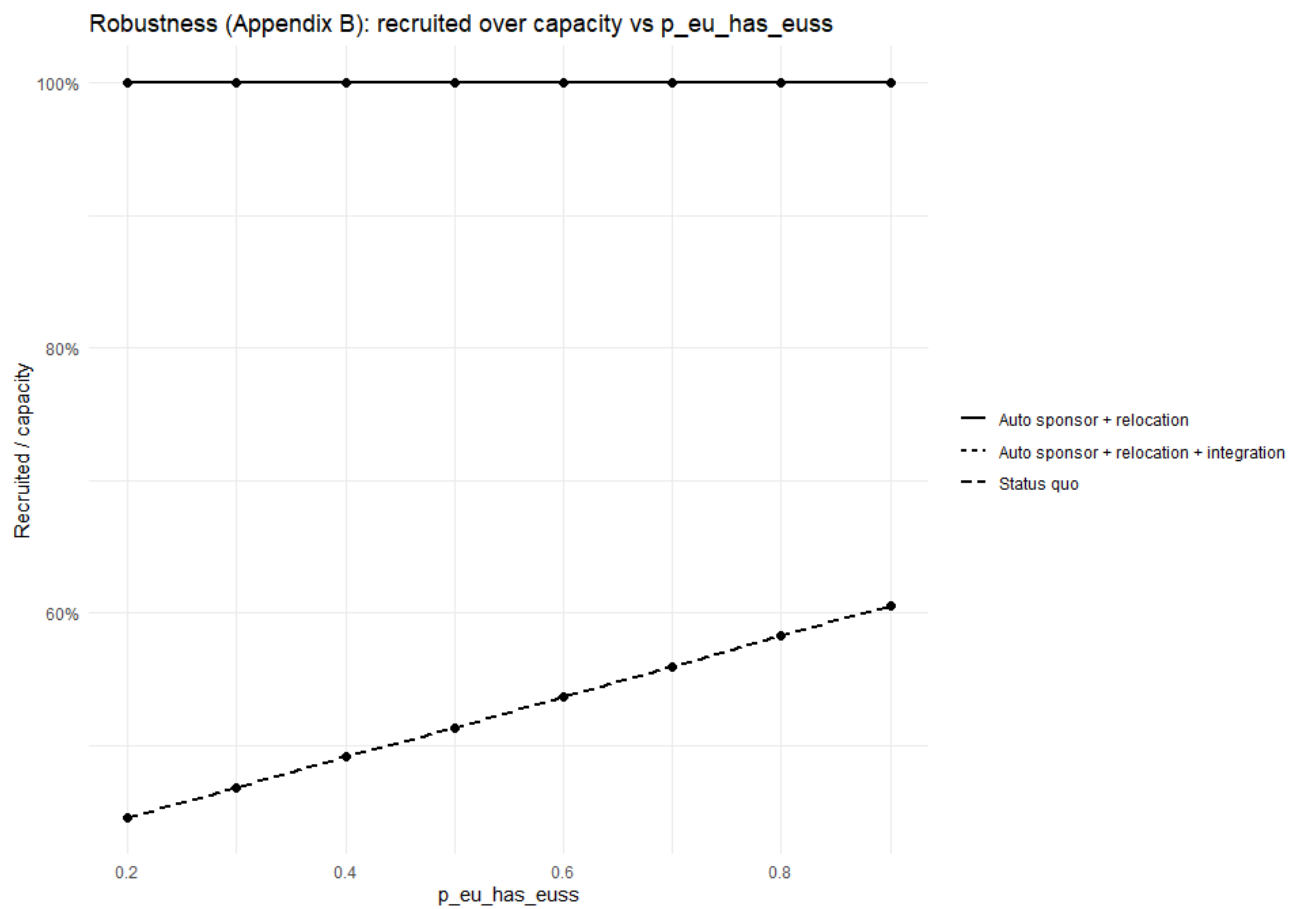


```

df <- robustB_means |>
  dplyr::filter(metric == "recruited_over_capacity")

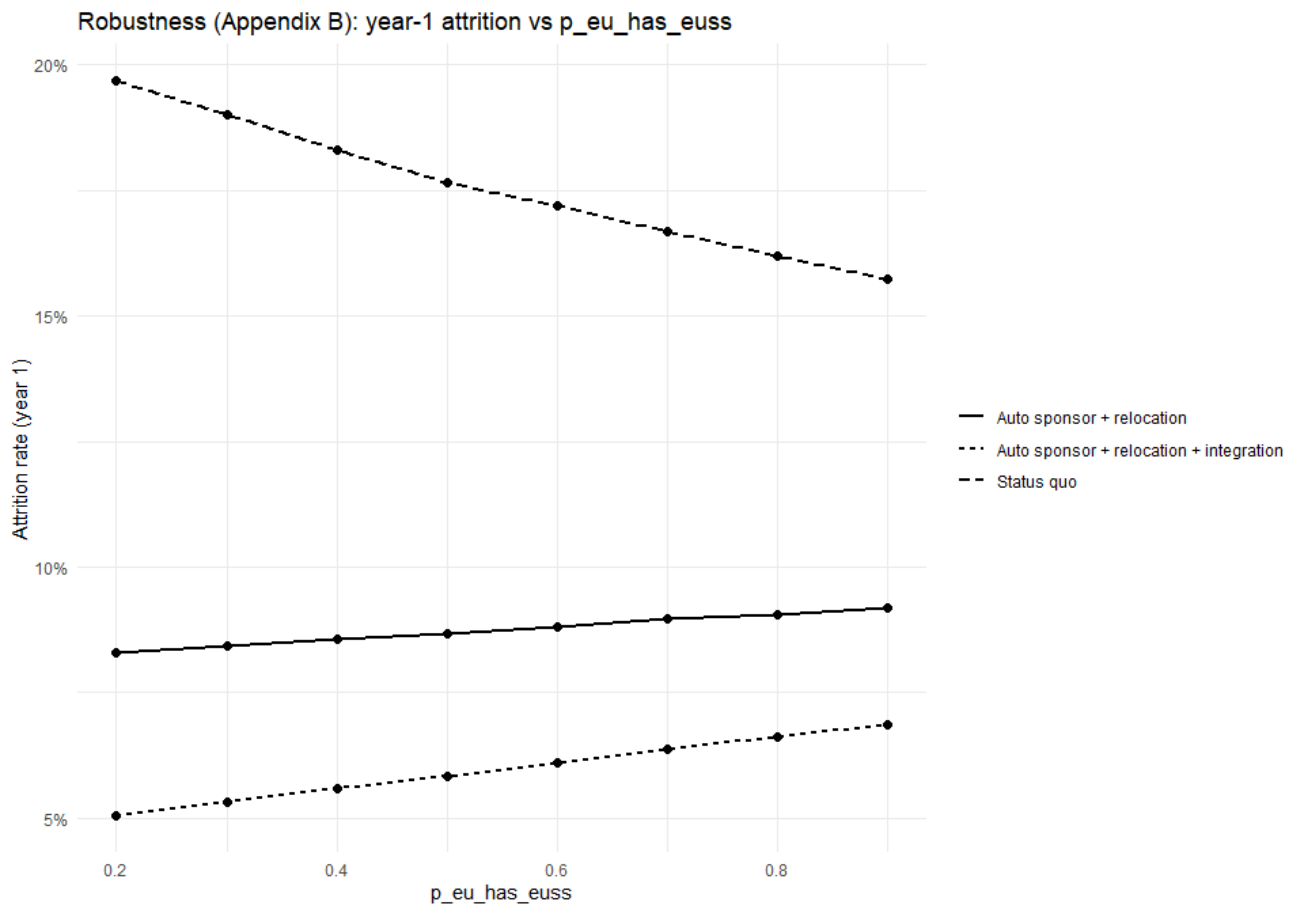
plot_line(
  df,
  ylab = "Recruited / capacity",
  title = "Robustness (Appendix B): recruited over capacity vs p_eu_has_euss",
  y_formatter = scales::percent_format(accuracy = 1)
)

```



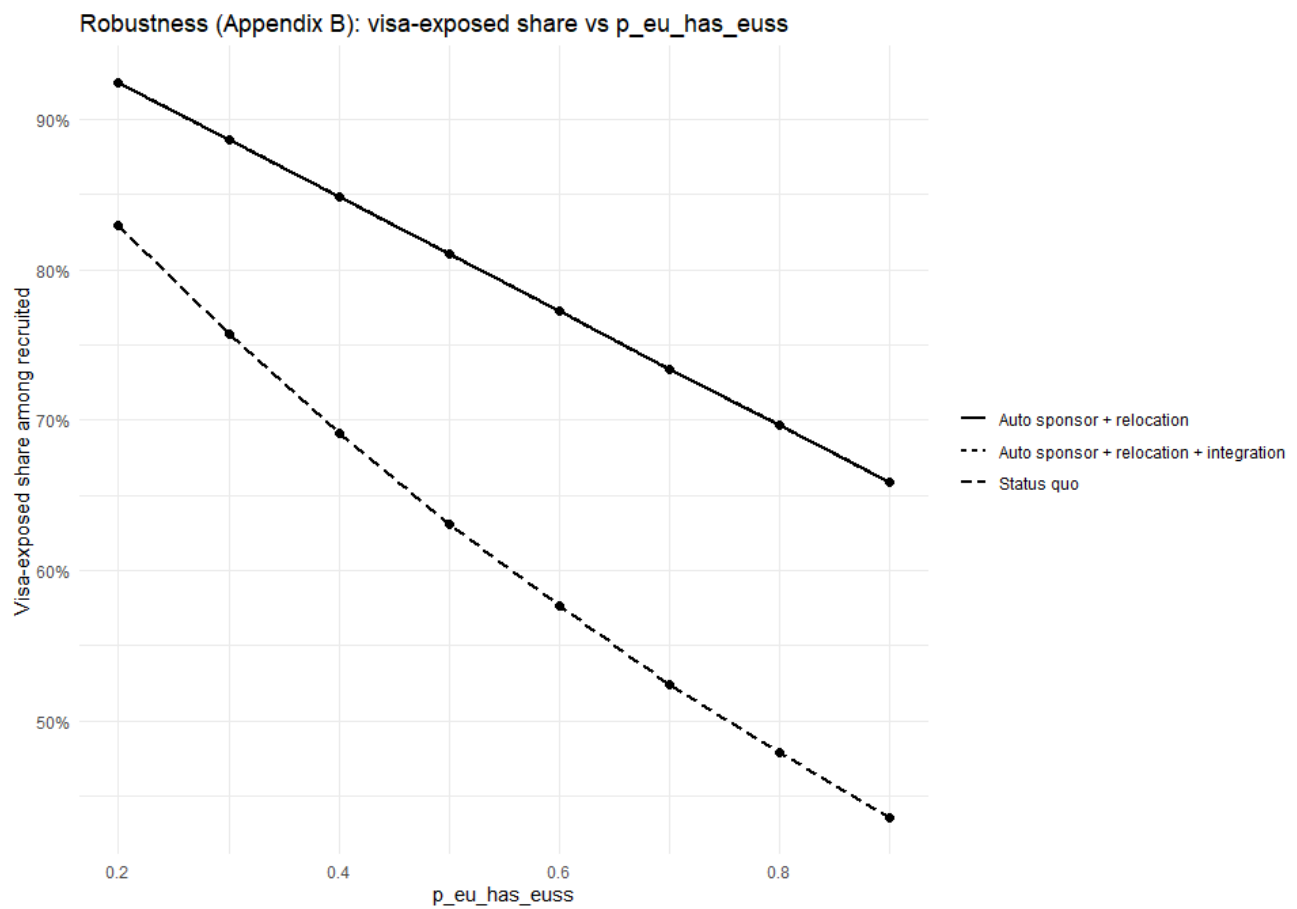
```
df <- robustB_means |>
  dplyr::filter(metric == "attrition_rate_year1")

plot_line(
  df,
  ylab = "Attrition rate (year 1)",
  title = "Robustness (Appendix B): year-1 attrition vs p_eu_has_euss",
  y_formatter = scales::percent_format(accuracy = 1)
)
```



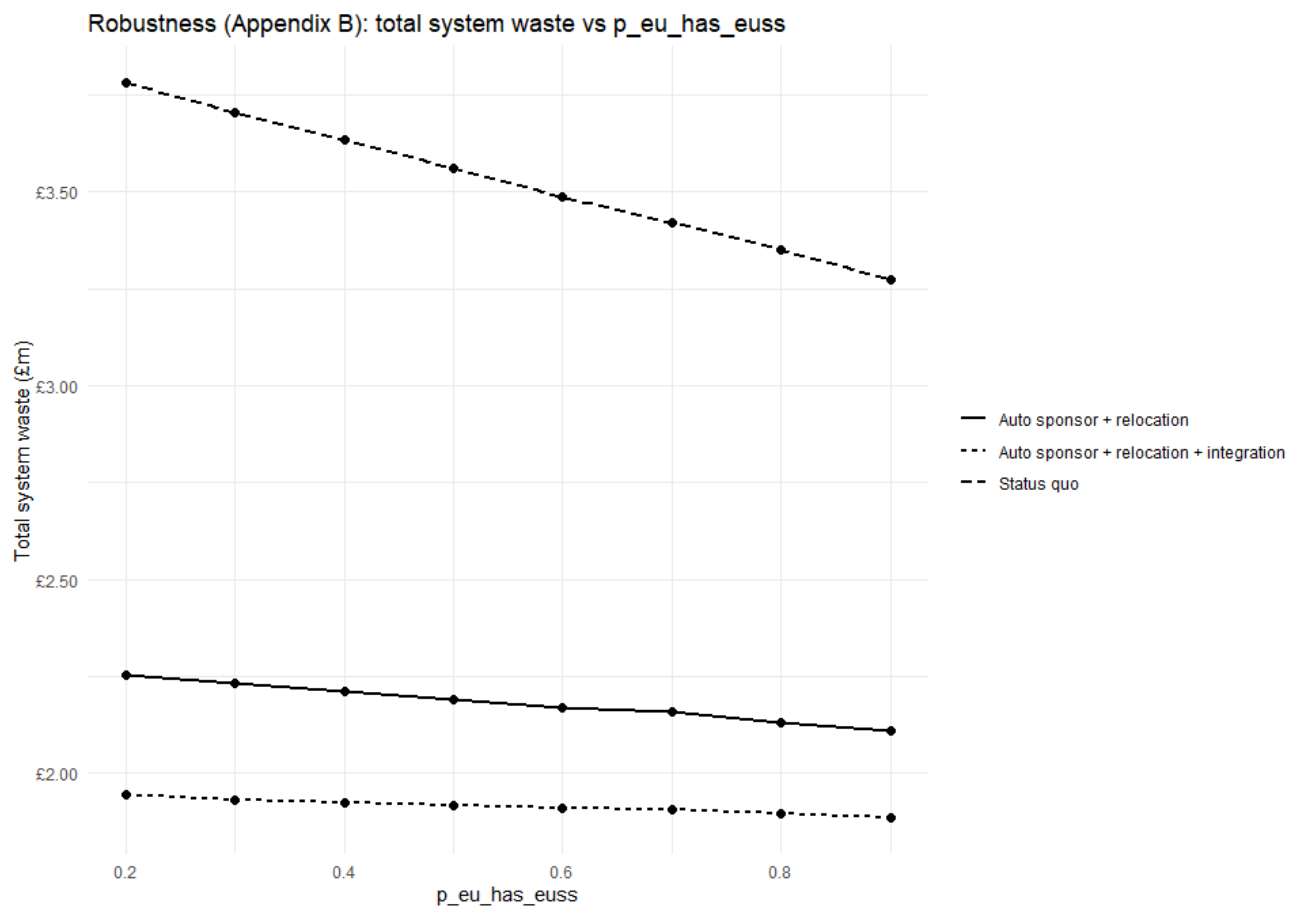
```
df <- robustB_means |>
  dplyr::filter(metric == "share_needs_visa_among_recruited")

plot_line(
  df,
  ylab = "Visa-exposed share among recruited",
  title = "Robustness (Appendix B): visa-exposed share vs p_eu_has_euss",
  y_formatter = scales::percent_format(accuracy = 1)
)
```



```
df <- robustB_means |>
  dplyr::filter(metric == "total_system_waste") |>
  dplyr::mutate(mean = mean / 1e6)

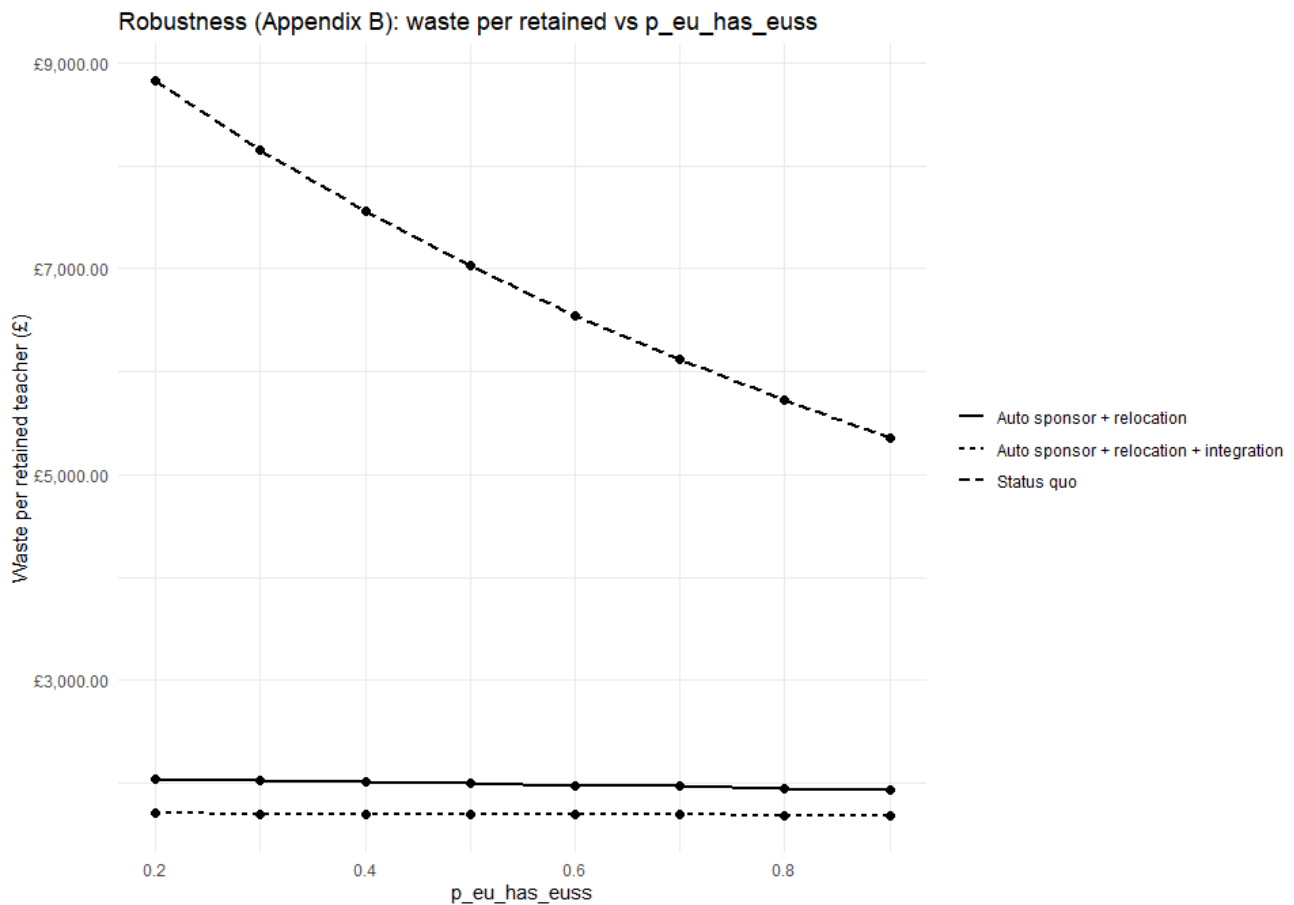
plot_line(
  df,
  ylab = "Total system waste (£m)",
  title = "Robustness (Appendix B): total system waste vs p_eu_has_euss",
  y_formatter = function(x) pound(x, accuracy = 0.01)
)
```



```
df <- robustB_means |>
  dplyr::filter(metric == "waste_per_retained")

plot_line(
  df,
  ylab = "Waste per retained teacher (£)",
  title = "Robustness (Appendix B): waste per retained vs p_eu_has_euss",
  y_formatter = function(x) pound(x, accuracy = 0.01)
)
```





```
# =====
# Appendix – Randomness & reproducibility checks (R = 5000 run)
# =====

R_target <- 5000L
mc_path <- cache_path("mc_abc_R5000")

need_cols <- c(
  "rep", "policy_id",
  "n_recruited", "attrition_rate_year1", "share_needs_visa_among_recruited",
  "total_dfe_waste", "total_school_waste", "total_teacher_waste",
  "total_system_waste", "waste_per_retained"
)

if (!csv_schema_matches(mc_path, need_cols)) {
  stop(
    "Missing MC cache or schema mismatch: ", mc_path, "\n",
    "Re-knit Block 4 to regenerate under the current cache_version."
  )
}

mc_abc <- readr::read_csv(mc_path, show_col_types = FALSE)

# -----
# A1) Integrity checks
```

```

# -----
rep_vals <- sort(unique(mc_abc$rep))

chk_rows <- (nrow(mc_abc) == 3L * R_target)
chk_policy_per_rep <- all(table(mc_abc$rep) == 3L)
chk_rep_range <- (min(rep_vals) == 1) && (max(rep_vals) == R_target) && (length(rep_v

integrity_tbl <- tibble::tibble(
  check = c("rows == 3*R", "3 policies per rep", "rep covers 1..R"),
  ok = c(chk_rows, chk_policy_per_rep, chk_rep_range)
)

# -----
# A2) Determinism check: rerun first K reps and compare
# -----
K_check <- 30L
if (!exists("run_one_replication")) stop("run_one_replication() not found (Block 4 mu

RNGkind("L'Ecuyer-CMRG")
set.seed(20250117)

rerun_list <- vector("list", K_check)
for (r in seq_len(K_check)) {
  rerun_list[[r]] <- run_one_replication(r, N0 = 2000, params = params)
}

keep_cols <- c(
  "rep", "policy_id",
  "n_recruited", "attrition_rate_year1", "share_needs_visa_among_recruited",
  "total_dfe_waste", "total_school_waste", "total_teacher_waste",
  "total_system_waste", "waste_per_retained"
)

mc_rerun <- dplyr::bind_rows(rerun_list) |>
  dplyr::select(dplyr::all_of(keep_cols)) |>
  dplyr::arrange(rep, policy_id) |>
  dplyr::mutate(dplyr::across(where(is.integer), as.double))

mc_saved <- mc_abc |>
  dplyr::filter(rep <= K_check) |>
  dplyr::select(dplyr::all_of(keep_cols)) |>
  dplyr::arrange(rep, policy_id) |>
  dplyr::mutate(dplyr::across(where(is.integer), as.double))

diff_mat <- abs(
  as.matrix(dplyr::select(mc_rerun, -rep, -policy_id)) -
  as.matrix(dplyr::select(mc_saved, -rep, -policy_id))
)

```

```

max_abs_diff <- max(diff_mat, na.rm = TRUE)
det_ok <- is.finite(max_abs_diff) && (max_abs_diff < 1e-12)

# -----
# A3) CRN effectiveness: paired vs unpaired variance
# -----
wide <- mc_abc |>
  dplyr::select(rep, policy_id, waste_per_retained, total_system_waste, attrition_rat
  tidyr::pivot_wider(
    names_from = policy_id,
    values_from = c(waste_per_retained, total_system_waste, attrition_rate_year1)
  )

d_paired <- wide |>
  dplyr::transmute(
    d_waste_BA = waste_per_retained_B - waste_per_retained_A,
    d_waste_CA = waste_per_retained_C - waste_per_retained_A,
    d_sys_BA = total_system_waste_B - total_system_waste_A,
    d_sys_CA = total_system_waste_C - total_system_waste_A
  )

set.seed(20250117)
perm <- sample.int(nrow(wide))

d_unpaired <- tibble::tibble(
  d_waste_BA = wide$waste_per_retained_B - wide$waste_per_retained_A[perm],
  d_waste_CA = wide$waste_per_retained_C - wide$waste_per_retained_A[perm],
  d_sys_BA = wide$total_system_waste_B - wide$total_system_waste_A[perm],
  d_sys_CA = wide$total_system_waste_C - wide$total_system_waste_A[perm]
)

var_ratio <- tibble::tibble(
  contrast = names(d_paired),
  var_paired = vapply(d_paired, stats::var, numeric(1), na.rm = TRUE),
  var_unpaired = vapply(d_unpaired, stats::var, numeric(1), na.rm = TRUE),
  ratio_unpaired_over_paired = var_unpaired / var_paired
)

checks_tbl <- dplyr::bind_rows(
  integrity_tbl,
  tibble::tibble(check = paste0("determinism: rerun matches saved (K=", K_check, ", m
  tibble::tibble(check = "CRN reduces variance (all ratios > 1)", ok = all(var_ratio$
)

print(checks_tbl)

```

```

## # A tibble: 5 × 2
##   check

```

ok

```
##   <chr>                                <lgl>
## 1 rows == 3*R                          TRUE
## 2 3 policies per rep                   TRUE
## 3 rep covers 1..R                      TRUE
## 4 determinism: rerun matches saved (K=30, max|Δ|<1e-12) FALSE
## 5 CRN reduces variance (all ratios > 1) TRUE
```

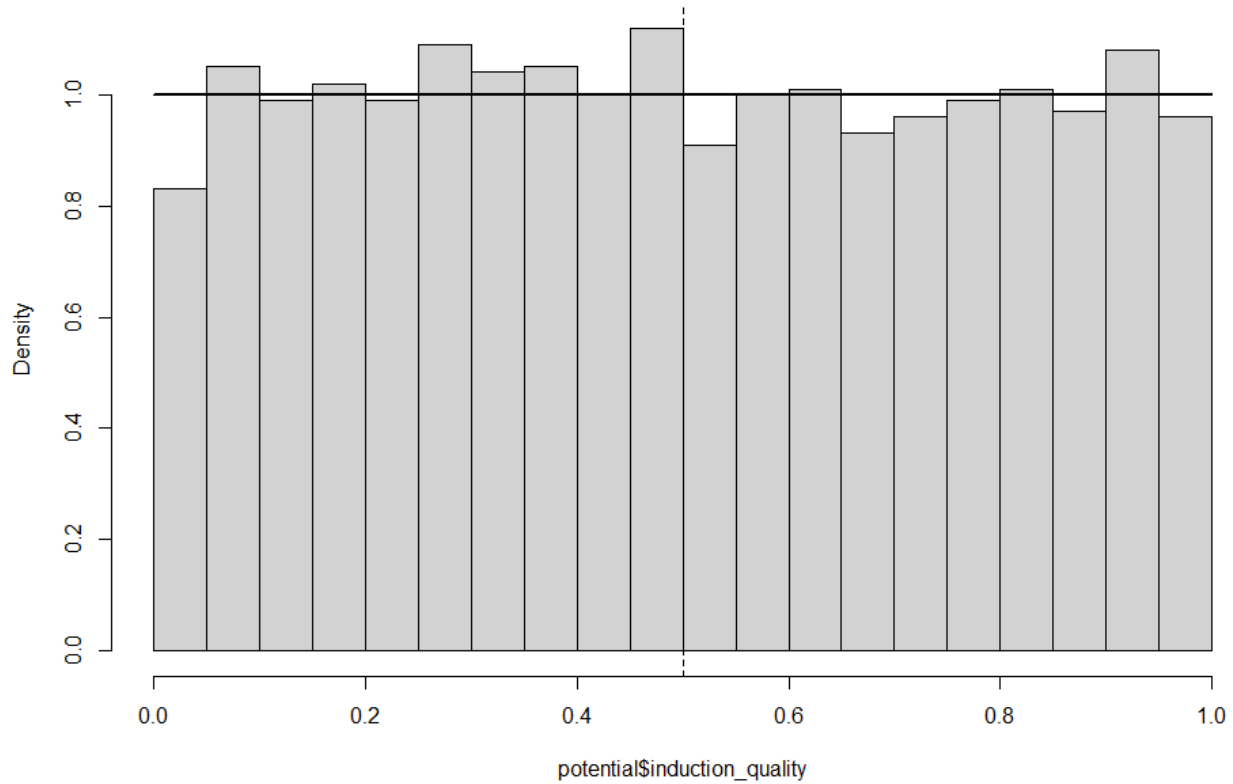
```
cat("Determinism max absolute difference:", signif(max_abs_diff, 6), "\n")
```

```
## Determinism max absolute difference: 890910
```

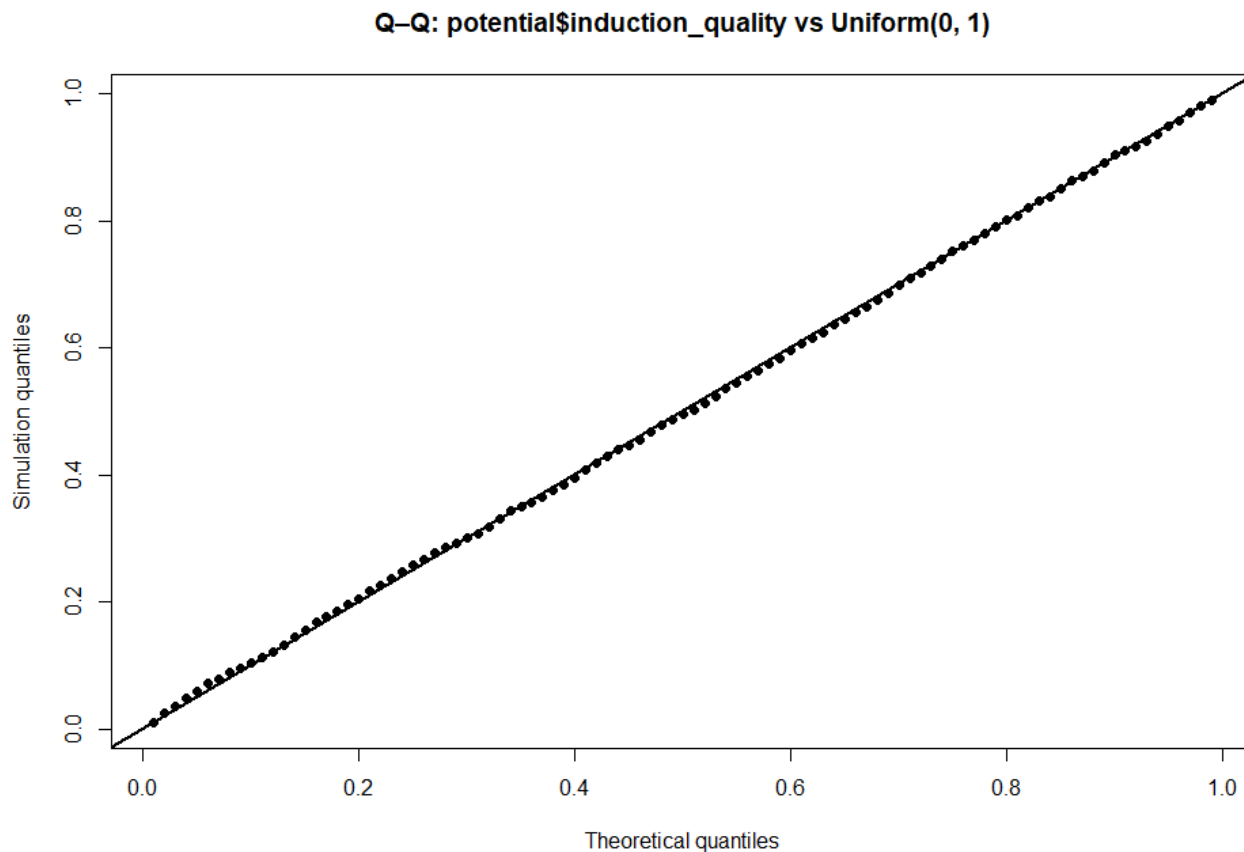
```
print(var_ratio)
```

```
## # A tibble: 4 × 4
##   contrast      var_paired var_unpaired ratio_unpaired_over_paired
##   <chr>          <dbl>         <dbl>                <dbl>
## 1 d_waste_BA    243397.         407356.                1.67
## 2 d_waste_CA    256222.         402162.                1.57
## 3 d_sys_BA     39427279479. 127606262115.         3.24
## 4 d_sys_CA     41619514143. 124417835716.         2.99
```

Visual test: potential\$induction\_quality ~ Uniform(0, 1)



```
##  
## [Moment matching] sample vs theory for potential$induction_quality  
  
## mean(sim) = 0.499988 | E[X] = 0.5 | abs diff = 1.22814e-05  
  
## var(sim) = 0.082082 | Var[X] = 0.0833333 | abs diff = 0.00125132
```



```
##
## [Q-Q tail check] abs deviation at probs {0.01,0.05,0.95,0.99}:

##      p=0.01      p=0.05      p=0.95      p=0.99
## 0.000532119 0.009727100 0.001547410 0.000244193
```

```
# =====
# Robustness post-check (NO re-simulation)
# Reads Appendix A cache and checks directional sanity.
# =====

stopifnot(exists("cache_path"), exists("csv_schema_matches"))
suppressPackageStartupMessages({
  library(dplyr); library(readr); library(tidyr)
})

robust_file <- cache_path("robust_p_eu_has_euss_R5000")

robust_required_cols <- c("p_eu_has_euss", "policy_id", "policy_short", "metric", "n", "me
if (!csv_schema_matches(robust_file, robust_required_cols)) {
  stop("Robustness cache missing or schema mismatch: ", robust_file)
}

robust <- readr::read_csv(robust_file, show_col_types = FALSE)
```

```

# Expected directional checks (rank-based, tolerant to small MC noise)
spearman_by <- robust %>%
  filter(metric %in% c(
    "share_needs_visa_among_recruited",
    "recruitment_rate",
    "total_teacher_waste",
    "waste_per_retained"
  )) %>%
  group_by(policy_id, policy_short, metric) %>%
  summarise(
    rho = suppressWarnings(cor(p_eu_has_euss, mean, method = "spearman")),
    .groups = "drop"
  )

# Rules of thumb (adapt if your narrative changes):
# - visa share should DECREASE as p_eu_has_euss increases => rho negative
visa_rho <- spearman_by %>%
  filter(metric == "share_needs_visa_among_recruited") %>%
  mutate(ok = is.finite(rho) & rho < -0.2)

# - recruitment_rate under Status Quo should generally INCREASE with p_eu_has_euss
recruit_rho <- spearman_by %>%
  filter(metric == "recruitment_rate") %>%
  mutate(ok = if_else(policy_id == "A", is.finite(rho) & rho > 0.2, TRUE))

# - teacher waste should generally DECREASE with p_eu_has_euss (less visa exposure)
teacher_waste_rho <- spearman_by %>%
  filter(metric == "total_teacher_waste") %>%
  mutate(ok = is.finite(rho) & rho < -0.1)

postcheck <- bind_rows(
  visa_rho %>% mutate(check = "Visa share decreases with p_eu_has_euss"),
  recruit_rho %>% mutate(check = "Recruitment (Status Quo) increases with p_eu_has_eu"),
  teacher_waste_rho %>% mutate(check = "Teacher waste decreases with p_eu_has_euss")
) %>%
  select(check, policy_id, policy_short, metric, rho, ok) %>%
  arrange(check, policy_id)

print(postcheck)

```

```

## # A tibble: 9 × 6
##   check                                policy_id policy_short metric    rho ok
##   <chr>                                <chr>      <chr>      <chr> <dbl> <lgl>
## 1 Recruitment (Status Quo) increases ... A        Status quo  recru...     1 TRUE
## 2 Recruitment (Status Quo) increases ... B        Auto sponso... recru...    NA TRUE
## 3 Recruitment (Status Quo) increases ... C        Auto sponso... recru...    NA TRUE
## 4 Teacher waste decreases with p_eu_h... A        Status quo   total...    -1 TRUE

```

|                                               |                         |         |
|-----------------------------------------------|-------------------------|---------|
| ## 5 Teacher waste decreases with p_eu_h... B | Auto sponso... total... | -1 TRUE |
| ## 6 Teacher waste decreases with p_eu_h... C | Auto sponso... total... | -1 TRUE |
| ## 7 Visa share decreases with p_eu_has_... A | Status quo share...     | -1 TRUE |
| ## 8 Visa share decreases with p_eu_has_... B | Auto sponso... share... | -1 TRUE |
| ## 9 Visa share decreases with p_eu_has_... C | Auto sponso... share... | -1 TRUE |

```

if (any(postcheck$ok == FALSE)) {
  warning("Some robustness directional checks failed. Inspect curves + MCSEs.")
} else {
  cat("✓ Robustness post-check: directional sanity looks consistent.\n")
}

```

```
## ✓ Robustness post-check: directional sanity looks consistent.
```